

Aromatic heterocycles 2: synthesis

30

Connections

➡ Building on

- Aromaticity [ch7](#)
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- Michael additions of enolates [ch25](#)
- Aldol reactions and acylation of enolates [ch26](#)
- Retrosynthetic analysis [ch28](#)
- Reactions of heterocycles [ch29](#)

Arriving at

- Thermodynamics is on our side
- Disconnecting the carbon–heteroatom bonds first
- How to make pyrroles, thiophenes, and furans from 1,4-dicarbonyl compounds
- How to make pyridines and pyridones
- How to make pyridazines and pyrazoles
- How to make pyrimidines from 1,3-dicarbonyl compounds and amidines
- How to make thiazoles
- How to make isoxazoles and tetrazoles by 1,3-dipolar cycloadditions
- The Fischer indole synthesis
- Making drugs: Viagra, sumatriptan, ondansetron, indomethacin
- How to make quinolines and isoquinolines

➡ Looking forward to

- Cycloadditions [ch34](#)
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In this chapter you will revisit the heterocyclic systems you have just met and find out how to make them. You'll also meet some new heterocyclic systems and find out how to make those. With so many heterocycles to consider, you'd be forgiven for feeling rather daunted by this prospect, but do not be alarmed. Making heterocycles is easy—that's precisely why there are so many of them. Just reflect...

- Making C–O, C–N, and C–S bonds is easy.
- Intramolecular reactions are preferred to intermolecular reactions.
- Forming five- and six-membered rings is easy.
- We are talking about aromatic, that is, very stable molecules.

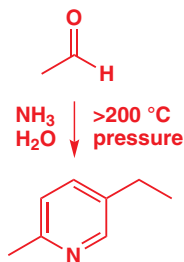
If we are to use these bullet points to our advantage we must think strategy before we start. When we were making benzene compounds we usually started with a preformed simple benzene derivative—toluene, phenol, aniline—and added side chains by electrophilic substitution. In this chapter our strategy will usually be to build the heterocyclic ring with most of its substituents already in place and add just a few others, perhaps by electrophilic substitution, but mostly by nucleophilic substitution.

We will usually make the rings by cyclization reactions with the heteroatom (O, N, S) as a nucleophile and a suitably functionalized carbon atom as the electrophile. This electrophile

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will almost always be a carbonyl compound of some sort and this chapter will help you revise your carbonyl chemistry from Chapters 10, 11, 20, 25, and 26 as well as the approach to synthesis described in Chapter 28.

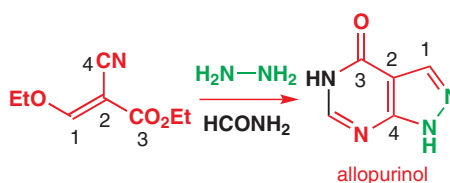
Thermodynamics is on our side



➡ Allopurinol was discussed in Chapter 29, p. 751.

Some of the syntheses we will meet will be quite surprisingly simple! It sometimes seems that we can just mix a few things together with about the right number of atoms and let thermodynamics do the rest. A commercial synthesis of pyridines combines acetaldehyde and ammonia under pressure to give a simple pyridine.

The yield is only about 50%, but what does that matter in such a simple process? By counting atoms we can guess that four molecules of aldehyde and one of ammonia react, but exactly how is a triumph of thermodynamics over mechanism. Much more complex molecules can sometimes be made very easily too. Take allopurinol, for example, which you met in the last chapter. It is not too difficult to work out where the atoms go—the hydrazine obviously gives rise to the pair of adjacent nitrogen atoms in the pyrazole ring and the ester group must be the origin of the carbonyl group (the colours and numbers illustrate this)—but would you have planned this synthesis?

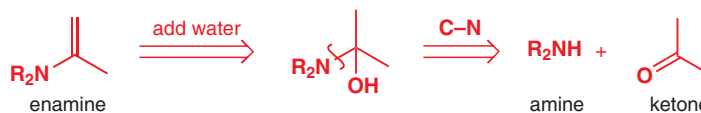


We will see that this sort of ‘witch’s brew’ approach to heterocyclic synthesis is restricted to a few basic ring systems and that, in general, careful planning is just as important here as elsewhere. The difference is that the synthesis of aromatic heterocycles is very forgiving—it often ‘goes right’ instead of going wrong. We’ll now look seriously at planning the synthesis of aromatic heterocycles.

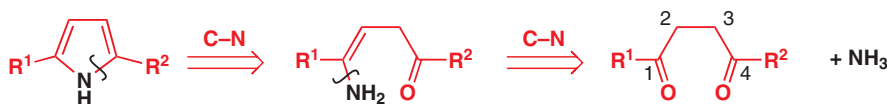
Disconnect the carbon–heteroatom bonds first

The simplest synthesis for a heterocycle emerges when we remove the heteroatom and see what electrophile we need. We shall use pyrroles as examples. The nitrogen forms an enamine on each side of the ring and we know that enamines are made from carbonyl compounds and amines.

➡ These arrows are the retrosynthetic arrows you met in Chapter 28.



If we do the same disconnection with a pyrrole, omitting the intermediate stage, we can repeat the C–N disconnection on the other side too:



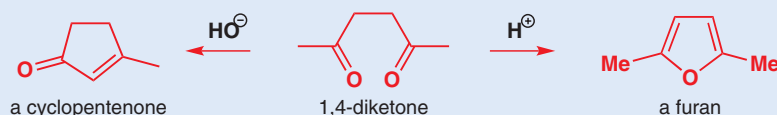
What we need is an amine—ammonia in this case—and a diketone. If the two carbonyl groups have a 1,4 relationship we will get a pyrrole out of this reaction. So hexane-2,5-dione reacts with ammonia to give a high yield of 2,5-dimethyl pyrrole. Making furans is even easier because the heteroatom (oxygen) is already there. All we have to do is to dehydrate the 1,4-diketone instead of making enamines from it. Heating with acid is enough.



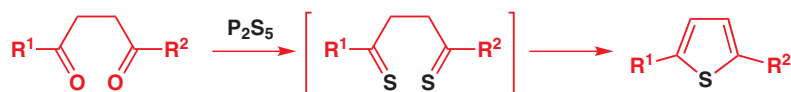
Avoiding the aldol product

1,4-Diketones also self-condense rather easily in an intramolecular aldol reaction (see Chapter 26, p. 636) to give a cyclopentenone with an all-carbon five-membered ring. This too is a useful reaction but we need to know how to control it. The usual rule is:

- Base gives the cyclopentenone.
- Acid gives the furan.



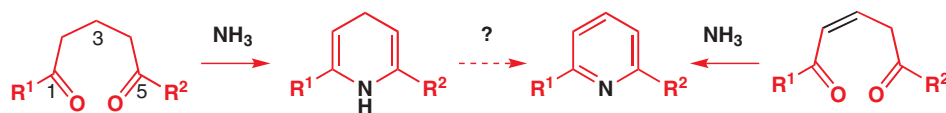
For thiophenes we could in theory use H_2S or some other sulfur nucleophile but, in practice, an electrophilic reagent is usually used to convert the two $\text{C}=\text{O}$ bonds to $\text{C}=\text{S}$ bonds. Thioketones are much less stable than ketones and cyclization is swift. Reagents such as P_2S_5 or Lawesson's reagent are the usual choice here.



• Making five-membered heterocycles

Cyclization of 1,4-dicarbonyl compounds with nitrogen, sulfur, or oxygen nucleophiles gives the five-membered aromatic heterocycles pyrrole, thiophene, and furan.

It seems a logical extension to use a 1,5-diketone to make substituted pyridines but there is a slight problem here as we will introduce only two of the required three double bonds when the two enamines are formed. To get the pyridine by enamine formation we should need a double bond somewhere in the chain between the two carbonyl groups. But here another difficulty arises—it will have to be a *cis* (*Z*) double bond or cyclization would be impossible.



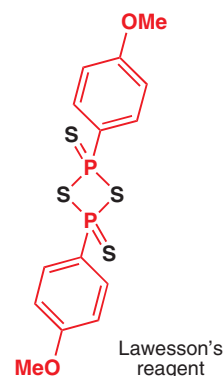
On the whole it is easier to use the saturated 1,5-diketone and oxidize the product to the pyridine. As we are going from a non-aromatic to an aromatic compound, oxidation is easy and we can replace the question mark above with almost any simple oxidizing agent, as we shall soon see.

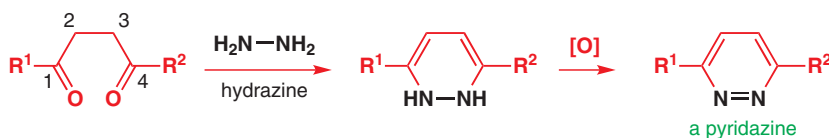
• Making six-membered heterocycles

Cyclization of 1,5-dicarbonyl compounds with nitrogen nucleophiles leads to the six-membered aromatic heterocycle pyridine.

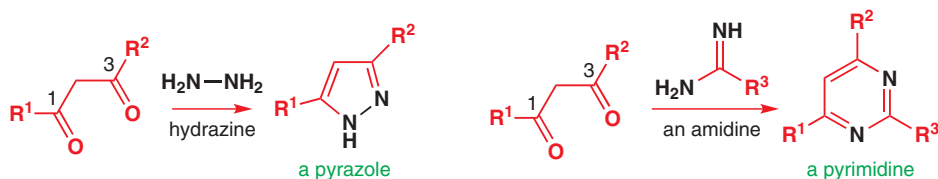
Heterocycles with two nitrogen atoms come from the same strategy

Reacting a 1,4-diketone with hydrazine (NH_2NH_2) makes a double enamine again and this is only an oxidation step away from a pyridazine. This is also a good synthesis.





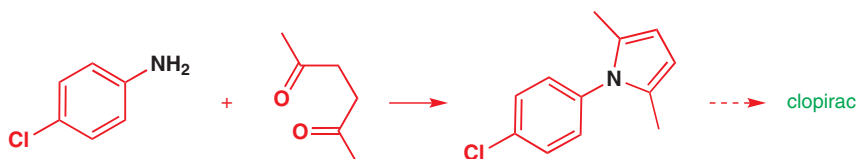
If we use a 1,3-diketone instead we will get a five-membered heterocycle and the imine and enamine formed are enough to give aromaticity without any need for oxidation. The product is a pyrazole. The two heteroatoms do not, of course, need to be joined together for this strategy to work. If an amidine is combined with the same 1,3-diketone we get a six-membered heterocycle. As the nucleophile contains one double bond already, an aromatic pyrimidine is formed directly.



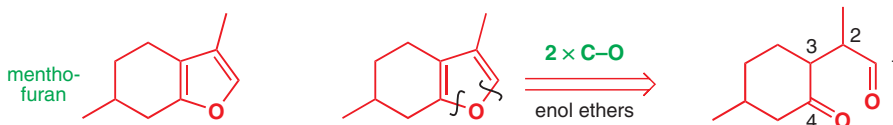
Since diketones and other dicarbonyl compounds are easily made by enolate chemistry (Chapters 25, 26, and 28) this strategy has been very popular and we will look at some detailed examples before moving on to more specialized reactions for the different classes of aromatic heterocycles.

Pyrroles, thiophenes, and furans from 1,4-dicarbonyl compounds

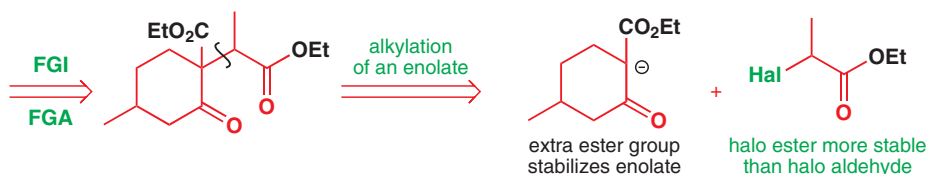
We need to make the point that pyrrole synthesis can be done with primary amines as well as with ammonia and a good example is the pyrrole needed for clopirac, a drug we discussed in Chapter 29. The synthesis is very easy.



For an example of furan synthesis we choose menthofuran, which contributes to the flavour of mint. It has a second ring, but that is no problem if we simply disconnect the enol ethers as we have been doing so far.

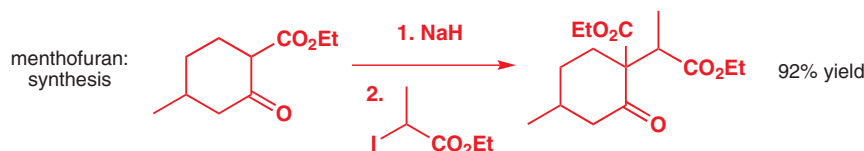


The starting material is again a 1,4-dicarbonyl compound but as there was no substituent at C1 of the furan, that atom is an aldehyde rather than a ketone. This might lead to problems in the synthesis so a few changes (using the notation you met in Chapter 28) are made to the intermediate before further disconnection.

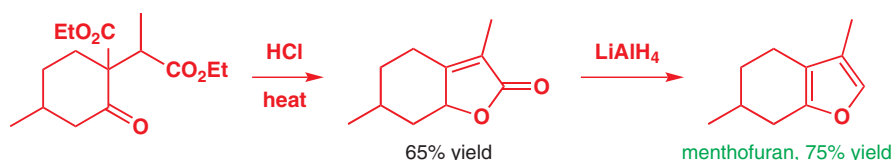


■ Halo aldehydes are unstable and should be avoided.

Notice in particular that we have ‘oxidized’ the aldehyde to an ester to make it more stable—in the synthesis reduction will be needed. Here is the alkylation step of the synthesis, which does indeed go very well with the α -iodo-ester.



Cyclization with acid now causes a lot to happen. The 1,4-dicarbonyl compound cyclizes to a lactone, not to a furan, and the redundant ester group is lost by hydrolysis and decarboxylation. Notice that the double bond moves into conjugation with the lactone carbonyl group. Finally, the reduction gives the furan. No special precautions are necessary—as soon as the ester is partly reduced, it loses water to give the furan whose aromaticity prevents further reduction even with LiAlH_4 .

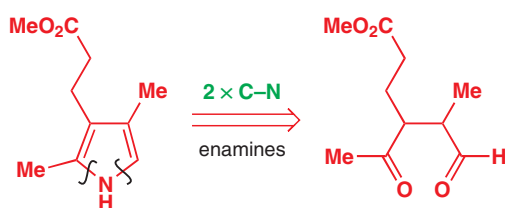


• A reminder

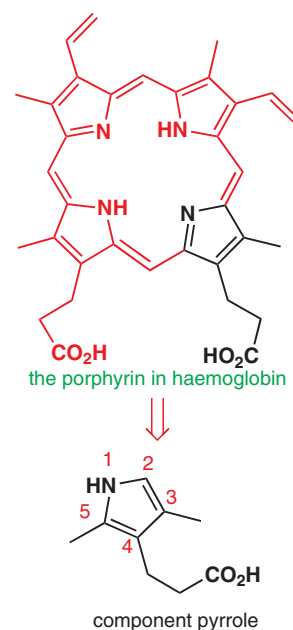
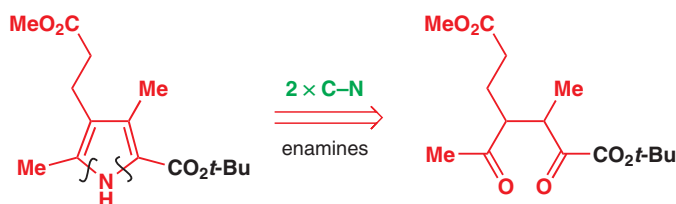
Cyclization of 1,4-dicarbonyl compounds with nitrogen, sulfur, or oxygen nucleophiles gives the five-membered aromatic heterocycles pyrrole, thiophene, and furan.

Now we need to take these ideas further and discuss an important pyrrole synthesis that follows this strategy but includes a cunning twist. It all starts with the porphyrin found in blood. In Chapter 29 we gave the structure of porphyrin and showed that it contains four pyrrole rings joined in a macrocycle. We are going to look at one of those pyrroles.

Porphyrins can be made by joining together the various pyrroles in the right order and what is needed for this one (and also, in fact, for another—the one in the north-east corner of the porphyrin) is a pyrrole with the correct substituents in positions 3 and 4, a methyl group in position 5, and a hydrogen atom at position 2. Position 2 must be free. Here is the molecule drawn somewhat more conveniently, together with the disconnection we have been using so far.

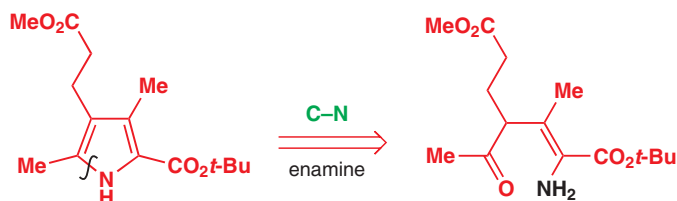


No doubt such a synthesis could be carried out but it is worth looking for alternatives for a number of reasons. We would prefer not to make a pyrrole with a free position at C2 as that would be very reactive and we know from Chapter 29 that we can reversibly block such a position with a *t*-butyl ester group. This gives us a very difficult starting material with four different carbonyl groups.

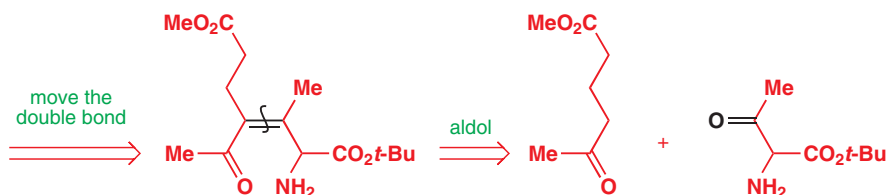


➡ See p. 733 for a discussion of how to control pyrrole's reactivity.

We have made a problem for ourselves by having two carbonyl groups next to each other. Could we escape from that by replacing one of them with an amine? We should then have an ester of an α amino acid, a much more attractive starting material, and this corresponds to disconnecting just one of the C–N bonds.

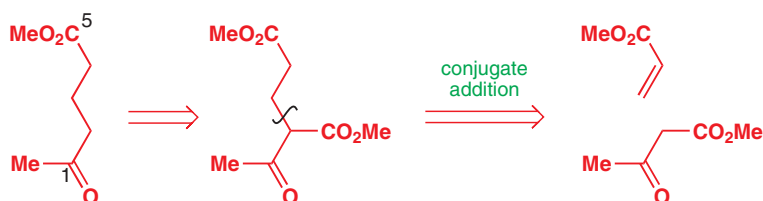


At first we seem to have made no progress but just see what happens when we move the double bond round the ring into conjugation with the ketone. After all, it doesn't matter where the double bond starts out—we will always get the aromatic product.

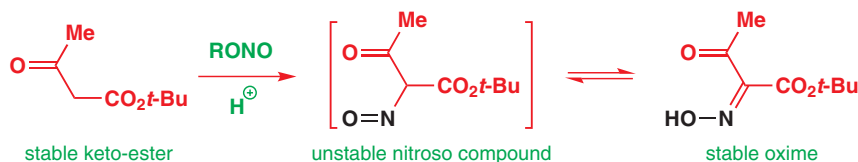


■ Conjugate additions with 1,3-dicarbonyl compounds were discussed in Chapter 26. If you have read Chapter 28 then you should be aware that such reactions are an excellent way of making 1,5-difunctionalized compounds.

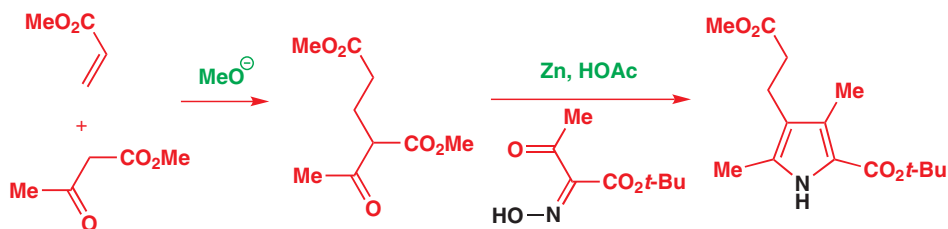
Each of our two much simpler starting materials needs to be made. The keto-ester is a 1,5-dicarbonyl compound so it can be made by a conjugate addition of an enolate, a process greatly assisted by the addition of a second ester group.



The other compound is an amino-keto-ester and will certainly react with itself if we try to prepare it as a pure compound. The answer is to release it directly into the reaction mixture and this can be done by nitrosation and reduction (Chapter 20) of another stable enolate.

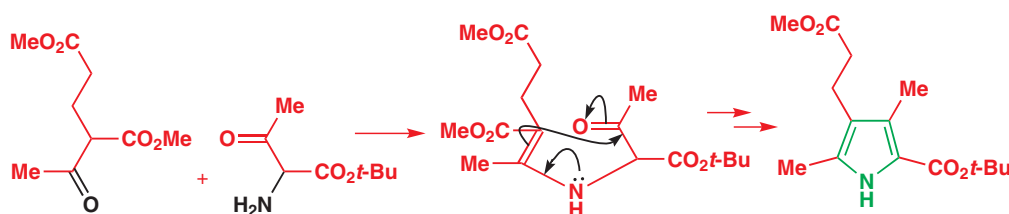


Zinc in acetic acid (Chapter 23) reduces the oxime to the amine and we can start the synthesis by doing the conjugate addition and then reducing the oxime in the presence of the keto-diester.

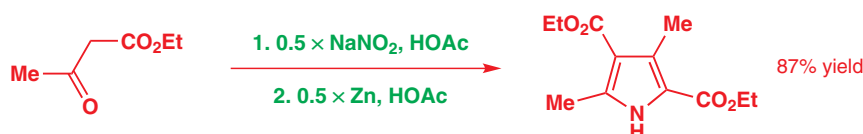


This reaction forms the required pyrrole in one step! First, the oxime is reduced to an amine, then the amino group forms an imine with the most reactive carbonyl group (the

ketone) in the ketodiester. Finally, the very easily formed enamine cyclizes onto the other ketone.



This pyrrole synthesis is important enough to be given the name of its inventor—it is the Knorr pyrrole synthesis. Knorr himself made a rather simpler pyrrole in a remarkably efficient reaction. See if you can work out what is happening here.

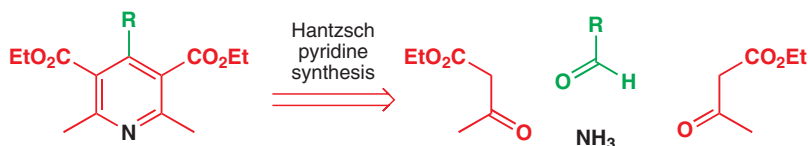


Names for heterocyclic syntheses

Standard heterocyclic syntheses tend to have a name associated with them and it is simply not worthwhile learning these names. Few chemists use any but the most famous of them: we will mention the Knorr pyrrole synthesis, the Hantzsch pyridine synthesis, and the Fischer and Reissert indole syntheses. We did not mention that the synthesis of furans from 1,4-dicarbonyl compounds is known as the Feist–Benary synthesis, and there are many more like this. If you are really interested in these other names we suggest you consult a specialist book on heterocyclic chemistry.

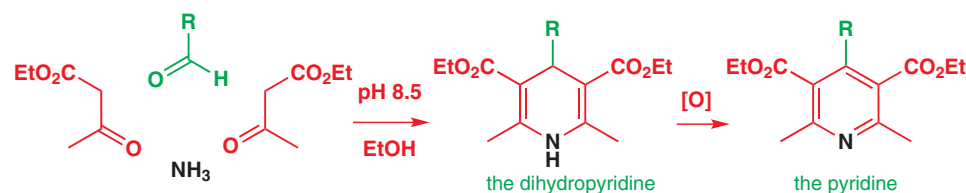
How to make pyridines: the Hantzsch pyridine synthesis

The idea of coupling two keto-esters together with a nitrogen atom also works for pyridines except that an extra carbon atom is needed. This is provided as an aldehyde and another important difference is that the nitrogen atom is added as a nucleophile rather than an electrophile. These are features of the **Hantzsch pyridine synthesis**. This is a four-component reaction from simple starting materials.



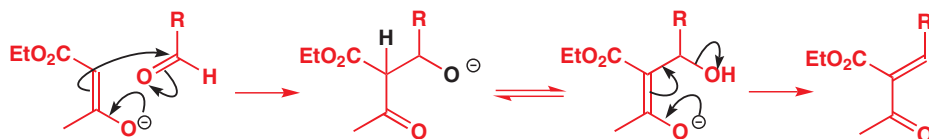
You are hardly likely to understand the rationale behind this reaction from that diagram so let's explore the details. The product of the reaction is actually the dihydropyridine, which has to be oxidized to the pyridine by a reagent such as HNO_3 , Ce(IV) , or a quinone.

The reaction is very simply carried out by mixing the components in the right proportions in ethanol. The presence of water does not spoil the reaction and the ammonia, or some added amine, ensures the slightly alkaline pH necessary. Any aldehyde can be used, even formaldehyde, and yields of the crystalline dihydropyridine are usually very good.

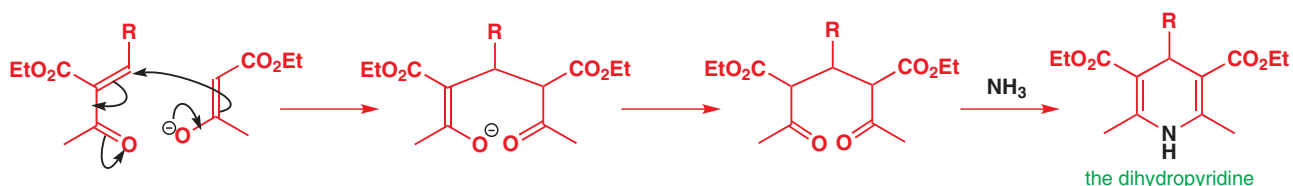


Arthur Hantzsch, 1857–1935, the 'fiery stereochemist' of Leipzig, was most famous for the work he did with Werner at the ETH in Zurich where in 1890 he suggested that oximes could exist in *cis* and *trans* forms.

This reaction is an impressive piece of molecular recognition by small molecules and writing a detailed mechanism is a bold venture. We can see that certain events have to happen, but which order they happen in is a matter of conjecture. The ammonia has to attack the ketone groups, but it would prefer to attack the more electrophilic aldehyde so this is probably not the first step. The enol or enolate of the keto-ester has to attack the aldehyde (twice!) so let us start there.

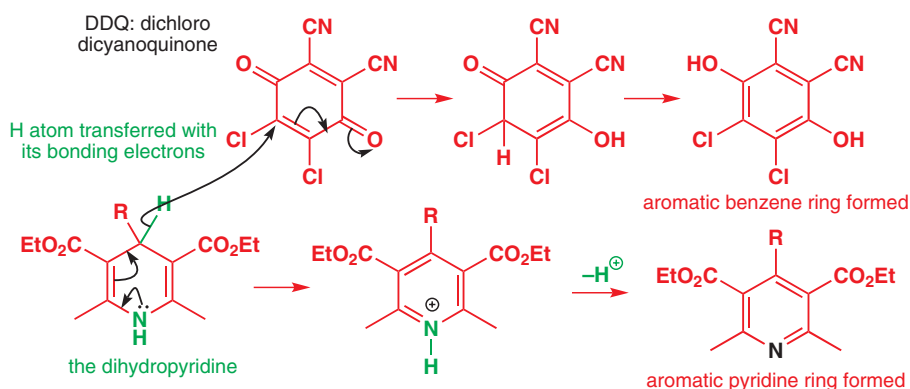


This adduct is in equilibrium with the stable enolate from the keto-ester and elimination now gives an unsaturated carbonyl compound. Such chemistry is associated with the aldol reactions we discussed in Chapter 26. The new enone has two carbonyl groups at one end of the double bond and is therefore a very good Michael acceptor (Chapter 25). A second molecule of enolate does a conjugate addition to complete the carbon skeleton of the molecule. Now the ammonia attacks either of the ketones and cyclizes on to the other. As ketones are more electrophilic than esters it is to be expected that ammonia will prefer to react there.



■ We will show in Chapter 42 that Nature uses related dihydropyridines as reducing agents in living things.

The necessary oxidation is easy both because the product is aromatic and because the nitrogen atom can help to expel the hydrogen atom and its pair of electrons from the 4-position. If we use a quinone as oxidizing agent, both compounds become aromatic in the same step.



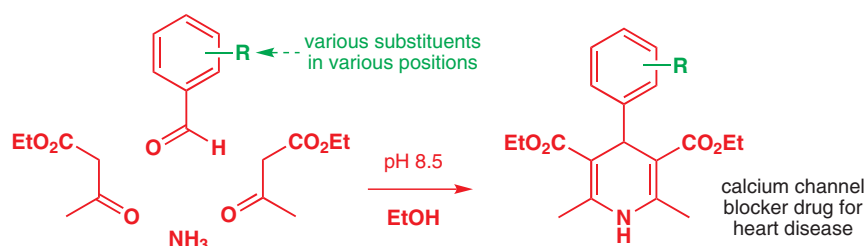
Interactive mechanism for quinone oxidation of dihydropyridines

To recap this mechanism, the essentials are:

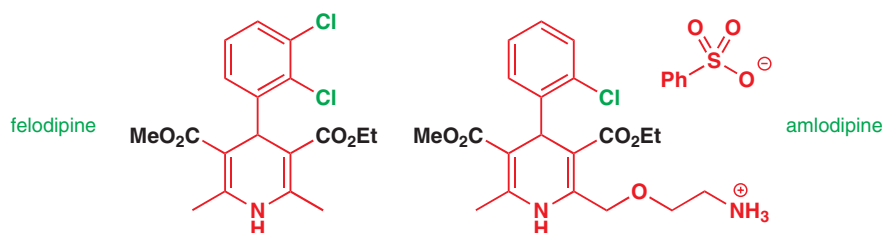
- aldol reaction between the aldehyde and the keto-ester
- Michael (conjugate) addition to the enone
- addition of ammonia to one ketone
- cyclization of the imine or enamine on to the other ketone

although several of the steps could happen in a different order.

The Hantzsch pyridine synthesis is an old discovery (1882) which sprang into prominence in the 1980s with the discovery that the dihydropyridine intermediates prepared from aromatic aldehydes are calcium channel-blocking agents and therefore valuable drugs for heart disease with useful effects on angina and hypertension.

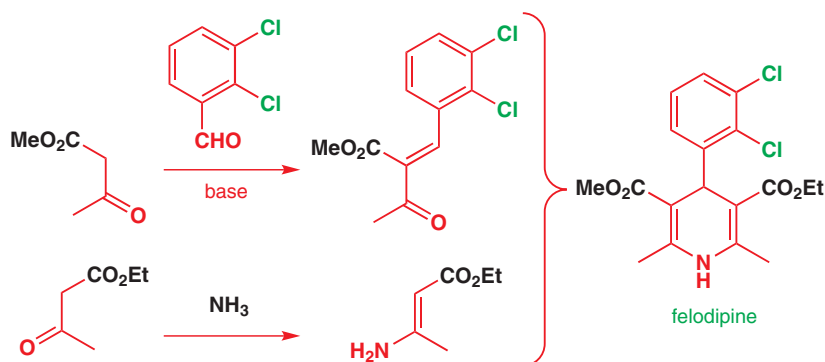


So far, so good. But it also became clear that the best drugs were unsymmetrical—some in a trivial way such as felodipine but some more seriously such as Pfizer's amlodipine. At first sight it looks as though the very simple and convenient Hantzsch synthesis cannot be used for these compounds.



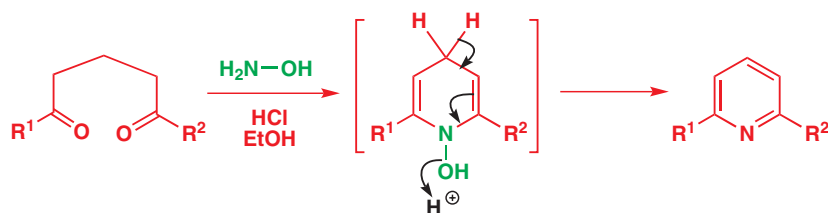
These drugs inhibit Ca^{2+} ion transport across cell membranes and relax muscle tissues selectively without affecting the working of the heart. They allow high blood pressure to be reduced. Pfizer's amlodipine (Istin™ or Norvasc™) is a very important drug.

Clearly, a modification is needed in which half of the molecule is assembled first. The solution lies in early work by Robinson, who made the very first enamines from keto-esters and amines. One half of the molecule is made from an enamine and the other half from a separately synthesized enone. We can use felodipine as a simple example.

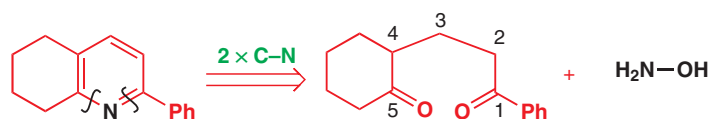


Other syntheses of pyridines

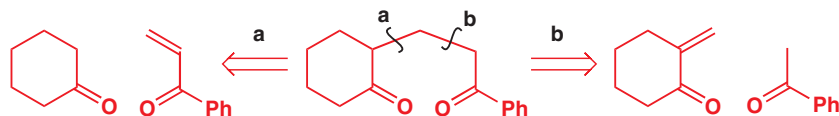
The Hantzsch synthesis produces a reduced pyridine but there are many syntheses that go directly to pyridines. One of the simplest is to use hydroxylamine (NH_2OH) instead of ammonia as the nucleophile. Reaction with a 1,5-diketone gives a dihydropyridine but then water is lost and no oxidation is needed.



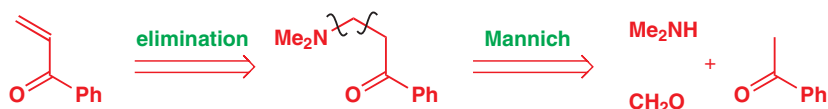
The example below shows how these 1,5-diketones may be quickly made by the Mannich (Chapter 26) and Michael (Chapter 25) reactions. Our pyridine has a phenyl substituent and a fused saturated ring. First we must disconnect to the 1,5-diketone.



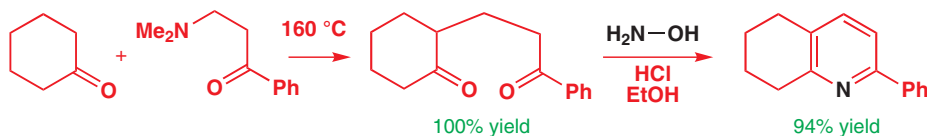
Further disconnection reveals a ketone and an enone. There is a choice here and both alternatives would work well.



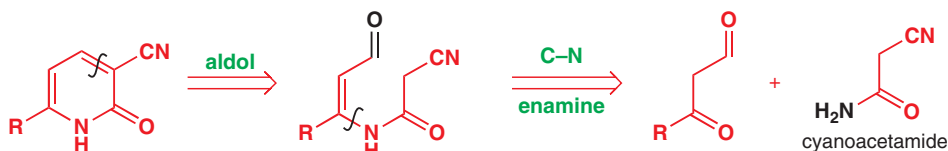
It is convenient to use the amine products of Mannich reactions ('Mannich bases') instead of the very reactive unsaturated ketones and we will continue with disconnection 'a'.



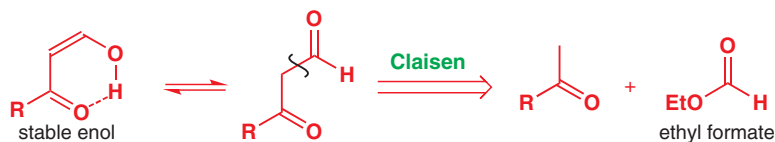
The synthesis is extraordinarily easy. The stable Mannich base is simply heated with the other ketone to give a high yield of the 1,5-diketone. Treatment of that with the HCl salt of NH_2OH in EtOH gives the pyridine directly, also in good yield.



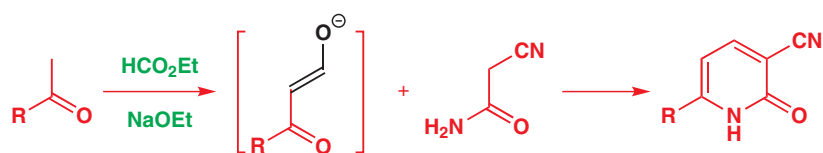
Another direct route leads, as we shall now demonstrate, to pyridones. These useful compounds are the basis for nucleophilic substitutions on the ring (Chapter 29). We choose an example that puts a nitrile in the 3-position. This is significant because the role of nicotinamide in living things (Chapter 42) makes such products interesting to make. Aldol disconnection of a 3-cyano pyridone starts us on the right path. If we now disconnect the C–N bond forming the enamine on the other side of the ring we will expose the true starting materials. This approach is unusual in that the nitrogen atom that is to be the pyridine nitrogen is not added as ammonia but is already present in a molecule of cyanoacetamide.



The keto-aldehyde can be made by a simple Claisen ester condensation (Chapter 26) using the enolate of the methyl ketone with ethyl formate (HCO_2Et) as the electrophile. It actually exists as a stable enol, like so many 1,3-dicarbonyl compounds (Chapter 20).

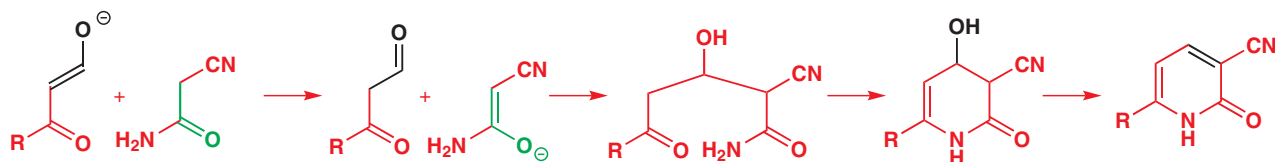


In the synthesis, the product of the Claisen ester condensation is actually the enolate anion of the keto-aldehyde and this can be combined directly without isolation with cyanoacetamide to give the pyridone in the same flask.



■ If dehydration occurred first, only the *Z* alkene could cyclize and the major product, the *E* alkene, would be wasted.

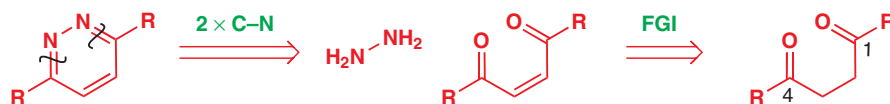
What happens here is that the two compounds must exchange protons (or switch enolates if you prefer) before the aldol reaction occurs. Cyclization probably occurs next through C–N bond formation and, finally, dehydration is forced to give the *Z* alkene.



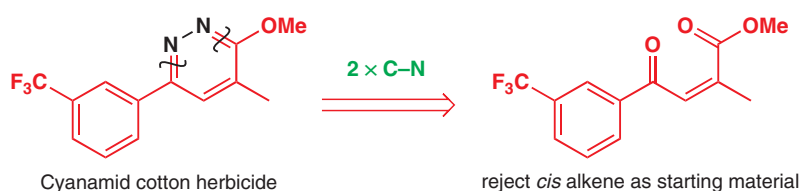
In planning the synthesis of a pyrrole or a pyridine from a dicarbonyl compound, considerable variation in oxidation state is possible. The oxidation state is chosen to make further disconnection of the carbon skeleton as easy as possible. We can now see how these same principles can be applied to pyrazoles and pyridazines.

Pyrazoles and pyridazines from hydrazine and dicarbonyl compounds

Disconnection of pyridazines reveals a molecule of hydrazine and a 1,4-diketone with the proviso that, just as with pyridines, the product will be a dihydropyridazine and oxidation will be needed to give the aromatic compound. As with pyridines, we prefer to avoid the *cis* double bond problem.

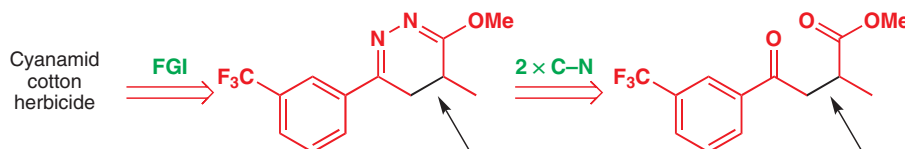


As an example we can take the cotton herbicide made by Cyanamid. Direct removal of hydrazine would require a problematic *cis* double bond in the starting material.

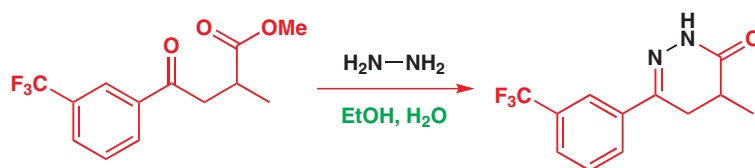


■ The herbicide kills weeds in cotton crops rather than the cotton plant itself!

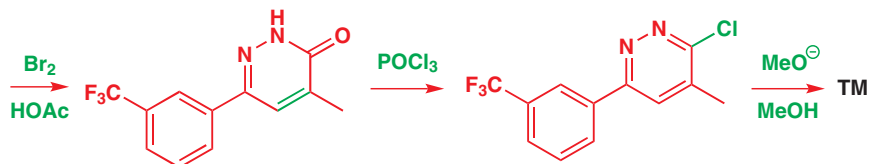
If we remove the double bond first, a much simpler compound emerges. Note that this is a ketoester rather than a diketone.



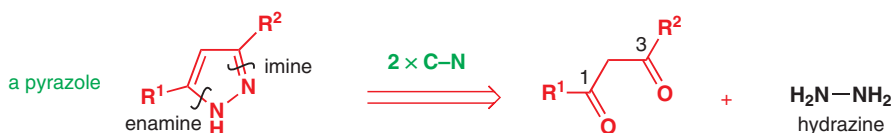
When hydrazine is added to the keto-ester an imine is formed with the ketone but acylation occurs at the ester end to give an amide rather than the imino-ester we had designed.



Aromatization with bromine gives the aromatic pyridazolone by bromination and dehydrobromination, and now we invoke the nucleophilic substitution reactions introduced in Chapter 29. First we make the chloride with POCl_3 and then displace with methanol.

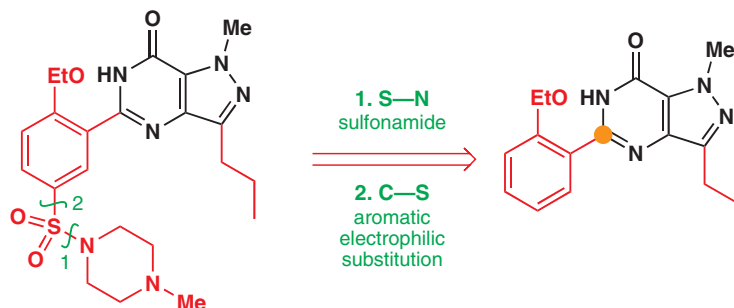
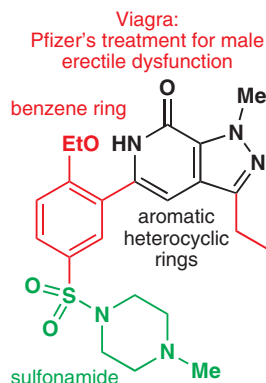


The five-membered ring pyrazoles are even simpler as the starting material is a 1,3-dicarbonyl compound available from the aldol or Claisen ester condensations.

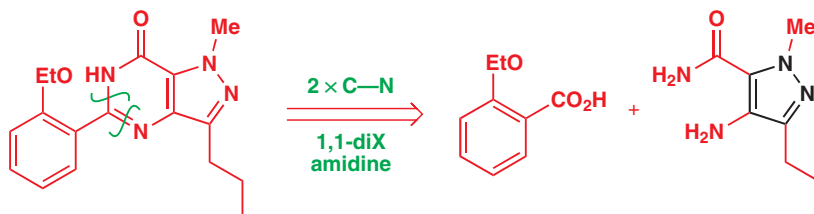


Chemistry hits the headlines—Viagra

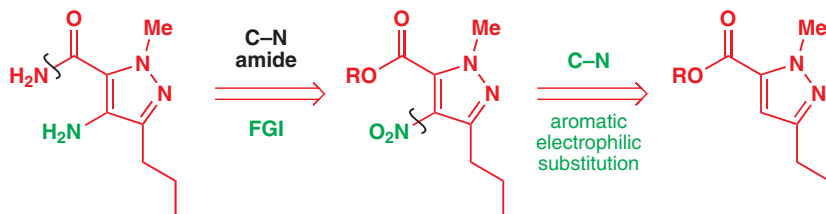
In 1998 chemistry suddenly appeared in the media in an exceptional way. Normally not a favourite of TV or the newspapers, chemistry produced a story with all the right ingredients—sex, romance, human ingenuity—and all because of a pyrazole. In the search for a heart drug, Pfizer uncovered a compound that allowed impotent men to have active sex lives. They called it Viagra. The molecule contains a sulfonamide and a benzene ring as well as the part that interests us most—a bicyclic aromatic heterocyclic system of a pyrazole fused to a pyrimidine. We shall discuss in detail how Pfizer made this part of the molecule and just sketch in the rest. The sulfonamide can be made from the sulfonic acid that can be added to the benzene ring by electrophilic aromatic sulfonation (Chapter 21).



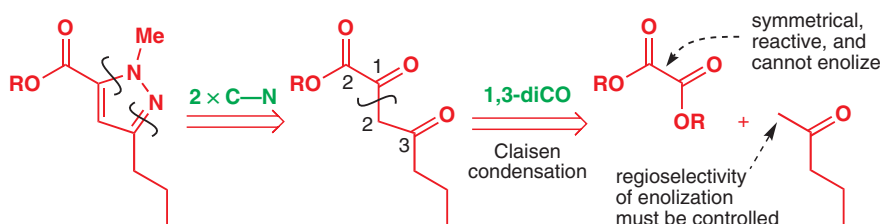
Inspection of what remains reveals that the carbon atom in the heterocycles next to the benzene ring (marked with an orange blob) is at the oxidation level of a carboxylic acid. If, therefore, we disconnect both $\text{C}-\text{N}$ bonds to this atom we will have two much simpler starting materials.



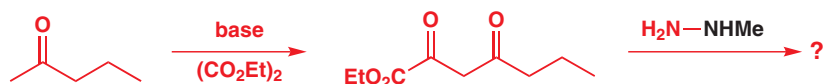
The aromatic acid is available and we need consider only the pyrazole (the core pyrazole ring in black in the diagram). The aromatic amino group can be put in by nitration and reduction, and the amide can be made from the corresponding ester. This leaves a carbon skeleton, which must be made by ring synthesis.



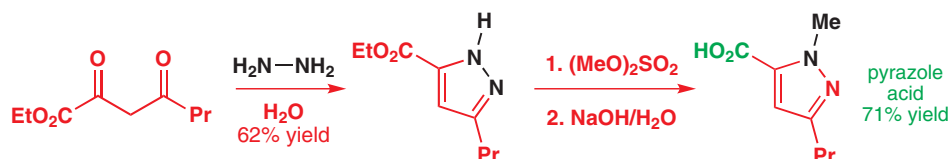
Following the methods we have established so far in this chapter, we can remove the hydrazine portion to reveal a 1,3-dicarbonyl compound. In fact, this is a tricarbonyl compound, a diketo-ester, because of the ester already present and it contains 1,2-, 1,3-, and 1,4-dicarbonyl relationships. The simplest synthesis is by a Claisen ester condensation and we choose the disconnection so that the electrophile is a reactive (oxalate) diester that cannot enolize. The only control needed will then be in the enolization of the ketone.



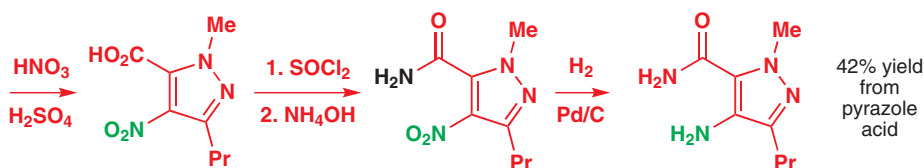
The Claisen ester condensation gives the right product just by treatment with base. The reasons for this are discussed in Chapter 26. We had then planned to treat the keto-diester with methylhydrazine but there is a doubt about the regioselectivity of this reaction—the ketones are more electrophilic than the ester all right, but which ketone will be attacked by which nitrogen atom?



We have already seen the solution to this problem in Chapter 29. If we use symmetrical hydrazine, we can deal with the selectivity problem by alkylation. Dimethyl sulfate turns out to be the best reagent.

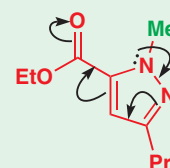


The stable pyrazole acid from the hydrolysis of this ester is a key intermediate in Viagra production. Nitration can occur only at the one remaining free position and then amide formation and reduction complete the synthesis of the amino pyrazole amide ready for assembly into Viagra.

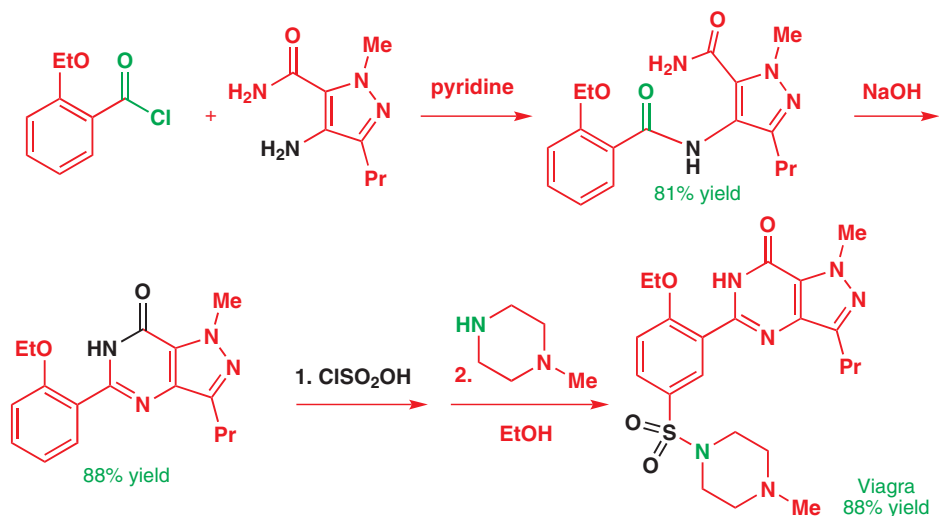


■ The alkylation is regioselective because the methylated nitrogen must become the pyrrole-like nitrogen atom and the molecule prefers the longest conjugated system involving that nitrogen and the ester.

lone pair delocalized into ester carbonyl group

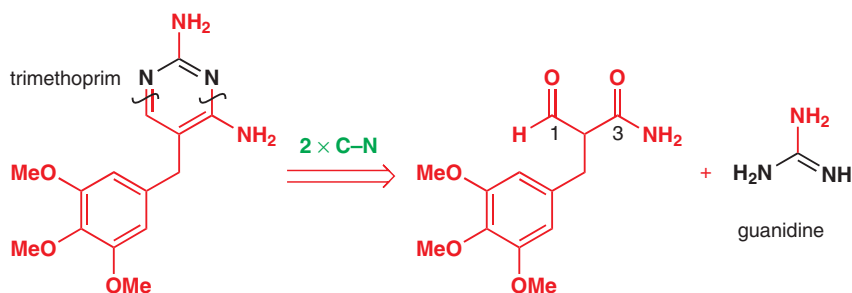


The rest of the synthesis can be summarized very briefly as it mostly concerns material outside the scope of this chapter. You might like to notice how easy the construction of the second heterocyclic ring is—the nucleophilic attack of the nitrogen atom of one amide on to the carbonyl of another would surely not occur unless the product were an aromatic heterocycle.

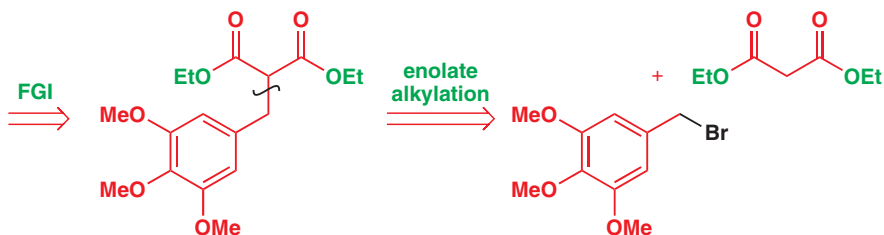


Pyrimidines can be made from 1,3-dicarbonyl compounds and amidines

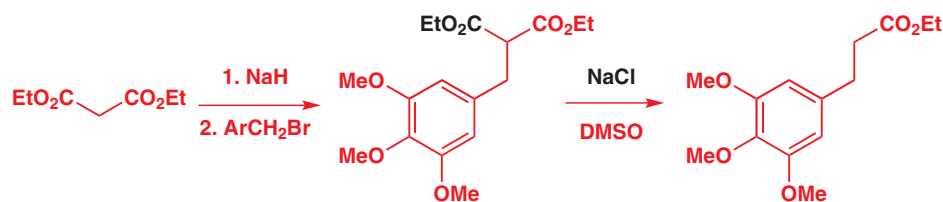
In Chapter 29 we met some compounds that interfere in folic acid metabolism and are used as antibacterial agents. One of them was trimethoprim and it contains a pyrimidine ring (black on the diagram). We are going to look at its synthesis briefly because the strategy used is the opposite of that used with the pyrimidine ring in Viagra. Here we disconnect a molecule of guanidine from a 1,3-dicarbonyl compound.



The 1,3-dicarbonyl compound is a combination of an aldehyde and an amide but is very similar to a malonic ester so we might think of making this compound by alkylation of that stable enolate (Chapter 25) with the convenient benzylic bromide.

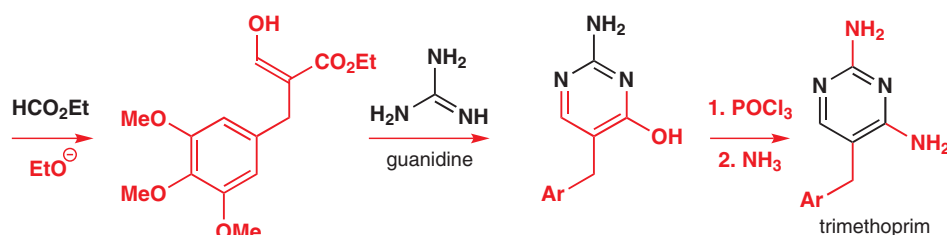


The alkylation works fine but it turns out to be better to add the aldehyde as an electrophile (cf. the pyridone synthesis on p. 766) rather than try to reduce an ester to an aldehyde. The other ester is already at the right oxidation level.



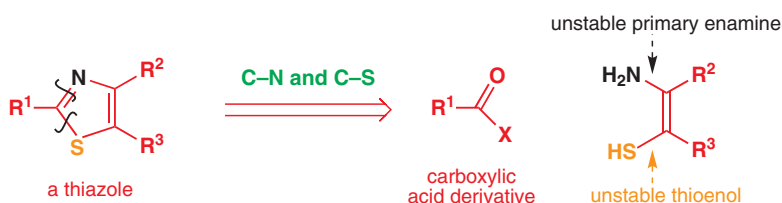
■ Notice the use of the NaCl method of decarboxylation (Chapter 25, p. 597).

Condensation with ethyl formate (HCO₂Et) and cyclization with guanidine gives the pyrimidine ring system but with an OH instead of the required amino group. Aromatic nucleophilic substitution the pyridone style from Chapter 29 gives trimethoprim.

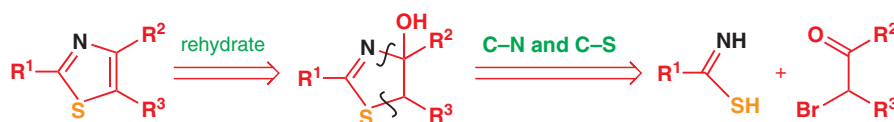


Unsymmetrical nucleophiles lead to selectivity questions

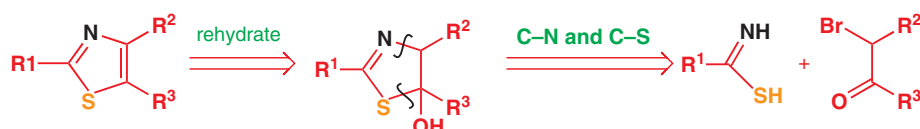
The synthesis of thiazoles is particularly interesting because of a regioselectivity problem. If we try out the two strategies we have just used for pyrimidines, the first requires the reaction of a carboxylic acid derivative with a most peculiar enamine that is also a thioenol. This does not look like a stable compound.



The alternative is to disconnect the C–N and C–S bonds on the other side of the heteroatoms. Here we must be careful what we are about or we will get the oxidation state wrong. We shall do it step by step to make sure. We can rehydrate the double bond in two ways. We can first try putting the OH group next to nitrogen.

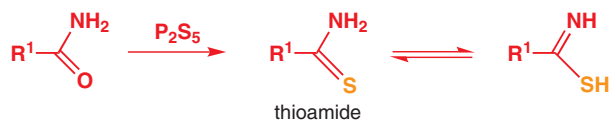


Or we can rehydrate it the other way round, putting the OH group next to the sulfur atom, and disconnect in the same way. In both cases we require an electrophilic carbon atom at the alcohol oxidation level and one at the aldehyde or ketone oxidation level. In other words we need an α-haloketone.

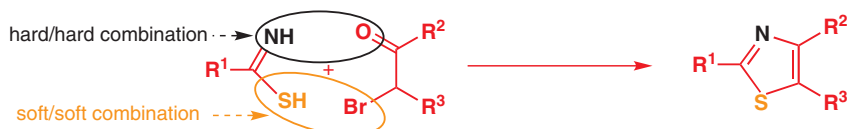


➡ The structure of Lawesson's reagent is on p. 759.

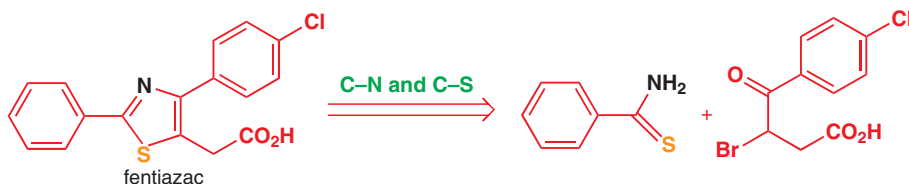
The nucleophile is the same in both cases and it is an odd-looking molecule. That is, until we realize that it is just a tautomer of a thioamide. Far from being odd, thioamides are among the few stable thiocarbonyl derivatives and can be easily made from ordinary amides with P_2S_5 or Lawesson's reagent.



So the only remaining question is: when thioamides combine with α -haloketones, which nucleophilic atom (N or S) attacks the ketone, and which atom (N or S) attacks the alkyl halide? Carbonyl groups are 'hard' electrophiles—their reactions are mainly under charge control and so they react best with basic nucleophiles (Chapter 10). Alkyl halides are 'soft' electrophiles—their reactions are mainly under frontier orbital control and they react best with large uncharged nucleophiles from the lower rows of the periodic table. The ketone reacts with nitrogen and the alkyl halide with sulfur.



Fentiazac, a non-steroidal anti-inflammatory drug, is a simple example. Disconnection shows that we need thiobenzamide and an easily made α -haloketone (easily made because the ketone can enolize on this side only—see Chapter 20).

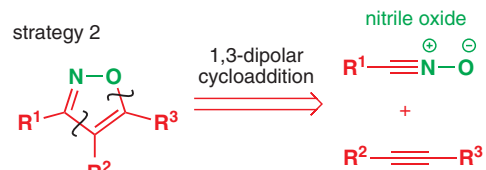
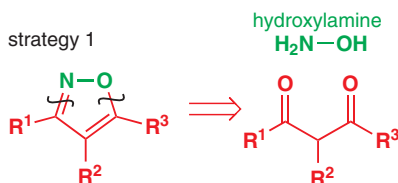
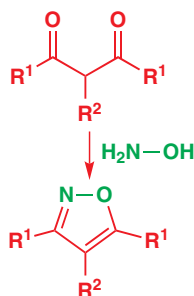


The synthesis involves heating these two compounds together and the correct thiazole forms easily with the double bonds finding their right positions in the product—the only positions for a stable aromatic heterocycle.

■ We will cover cycloadditions in detail in Chapter 34—in fact you have met an example already in the last chapter: the Diels–Alder reaction (p. 739) is also a cycloaddition. The cycloadditions here involve a 1,3-dipole and form a five-membered ring, by a cyclic mechanism involving six electrons.

Isoxazoles are made from hydroxylamine or by cycloaddition

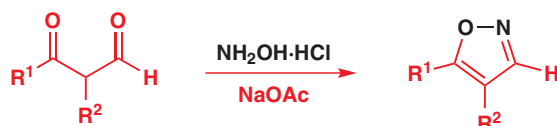
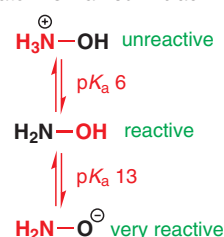
The two main routes for the synthesis of isoxazoles are (a) the attack of hydroxylamine (NH_2OH) on diketones and (b) a reaction of nitrile oxides called a 1,3-dipolar cycloaddition. They thus form a link between the strategy we have been discussing (cyclization of a nucleophile with two heteroatoms and a compound with two electrophilic carbon atoms) and the next strategy—cycloaddition reactions.



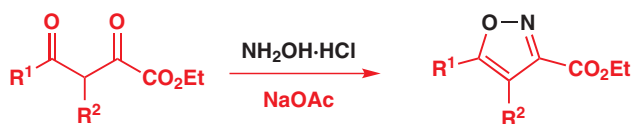
Simple symmetrical isoxazoles are easily made by the hydroxylamine route. If $R^1 = R^3$, we have a symmetrical and easily prepared 1,3-diketone as starting material. The central R^2 group can be inserted by alkylation of the stable enolate of the diketone (Chapter 25).

When $R^1 \neq R^3$, we have an unsymmetrical dicarbonyl compound and we must be sure that we know which way round the reaction will proceed. The more nucleophilic end of NH_2OH will attack the more electrophilic carbonyl group. It seems obvious that the more nucleophilic end of NH_2OH will be the nitrogen atom but that depends on the pH of the solution. Normally, hydroxylamine is supplied as the crystalline hydrochloride salt and a base of some kind added to give the nucleophile. The relevant pK_a s are shown in the margin. Bases such as pyridine or sodium acetate produce some of the reactive neutral NH_2OH in the presence of the less reactive cation, but bases such as NaOEt produce the anion. Reactions of keto-aldehydes with acetate-buffered hydroxylamine usually give the isoxazole from nitrogen attack on the aldehyde as expected.

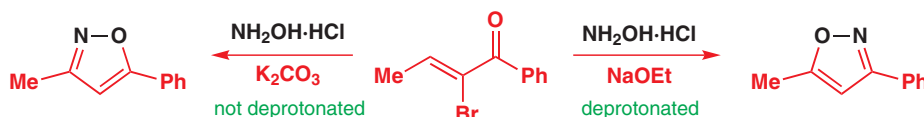
state of hydroxylamine changes with pH: the more nucleophilic atom is marked in black



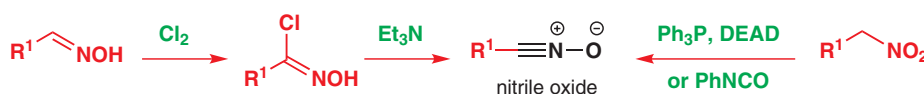
Modification of the electrophile may also be successful. Reaction of hydroxylamine with 1,2,4-diketo-esters usually gives the isoxazole from attack of nitrogen at the more reactive keto group next to the ester.



A clear demonstration of selectivity comes from the reactions of bromoenones. It is not immediately clear which end of the electrophile is more reactive but the reactions tell us the answer.

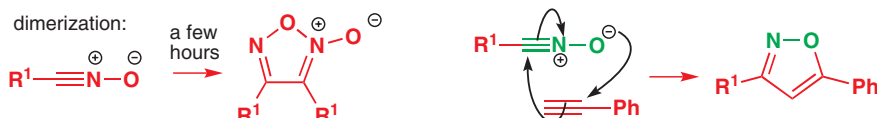


The alternative approach to isoxazoles relies on the reaction of nitrile oxides with alkynes. We shall see in Chapter 34 that there are two good routes to these reactive compounds, the γ -elimination of chlorooximes or the dehydration of nitroalkanes.



Interactive mechanism for nitrile oxide formation

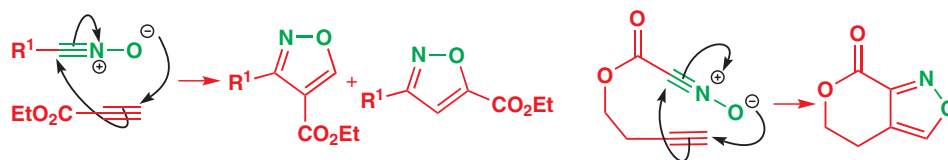
A few nitrile oxides are stable enough to be isolated (those with electron-withdrawing or highly conjugating substituents, for example) but most are prepared in the presence of the alkyne by one of these methods because otherwise they dimerize rapidly. Both methods of forming nitrile oxides are compatible with their rapid reactions with alkynes. With aryl alkynes the reaction is usually clean and regioselective.



Interactive mechanism for nitrile oxide cycloaddition

The reaction forms the five-membered ring in a single step: it is a cycloaddition, in which the alkyne is using its HOMO to attack the LUMO of the nitrile oxide (see Chapter 34 for an explanation). If the alkyne has an electron-withdrawing group, mixtures of isomers are usually formed as the HOMO of the nitrile oxide also attacks the LUMO of the alkyne. Intramolecular reactions are usually clean regardless of the preferred electronic orientation if

the tether is too short to allow any cyclization except one. In this example, even the more favourable orientation looks very bad because of the linear nature of the reacting species, but only one isomer is formed.

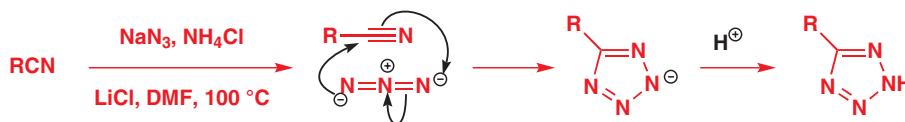


Tetrazoles and triazoles are also made by cycloadditions

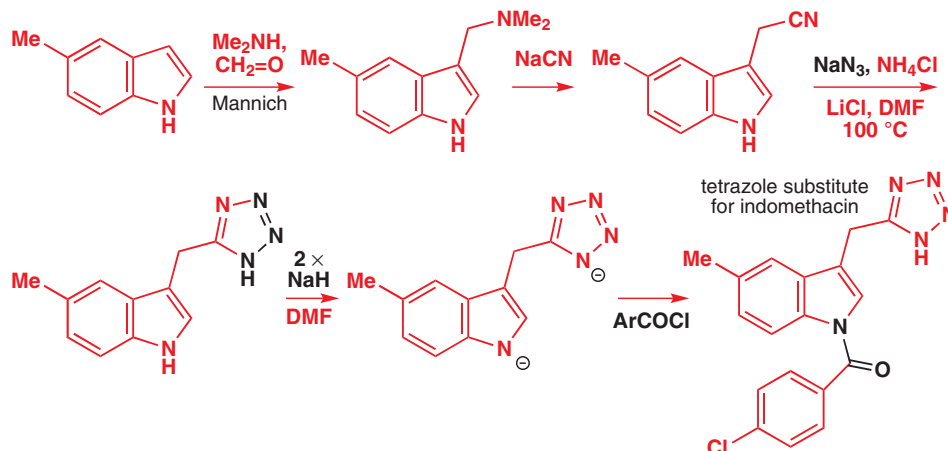
Disconnection of tetrazoles with a 1,3-dipolar cycloaddition in mind is easy to see once we realize that a nitrile (RCN) is going to be one of the components. It can be done in two ways: disconnection of the neutral compound would require the dangerous hydrazoic acid (HN₃) as the dipole but the anion disconnects directly to azide ion.



Unpromising though this reaction may look, it actually works well if an ammonium chloride-buffered mixture of sodium azide and the nitrile is heated in DMF. The reagent is really ammonium azide and the reaction occurs faster with electron-withdrawing substituents in R. In the reaction mixture, the anion of the tetrazole is formed but neutralization with acid gives the free tetrazole.



As nitriles are generally readily available this is the main route to simple tetrazoles. More complicated ones are made by alkylation of the product of a cycloaddition. The tetrazole substitute for indomethacin that we mentioned in Chapter 29 is made by this approach. First, the nitrile is prepared from the indole. The 1,3-dipolar cycloaddition works well by the azide route we have just discussed, even though this nitrile will form an 'enol' rather easily. Finally, the indole nitrogen atom must be acylated. The tetrazole is more acidic so it is necessary to form a dianion to get reaction at the right place. The usual rule is followed (see Chapter 23)—the second anion formed is less stable and so it reacts first.



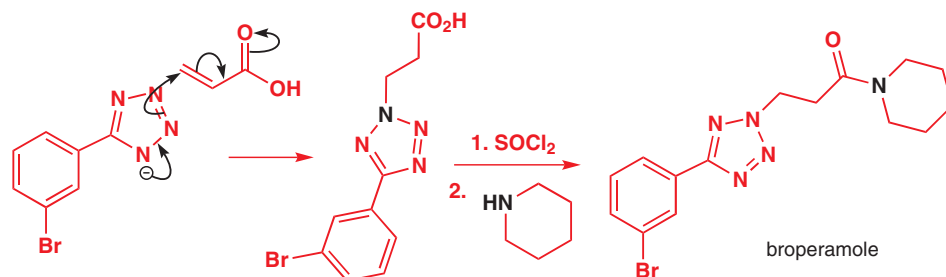
■ You saw in Chapter 29 that tetrazoles are about as acidic as carboxylic acids.

Interactive mechanism for tetrazole formation

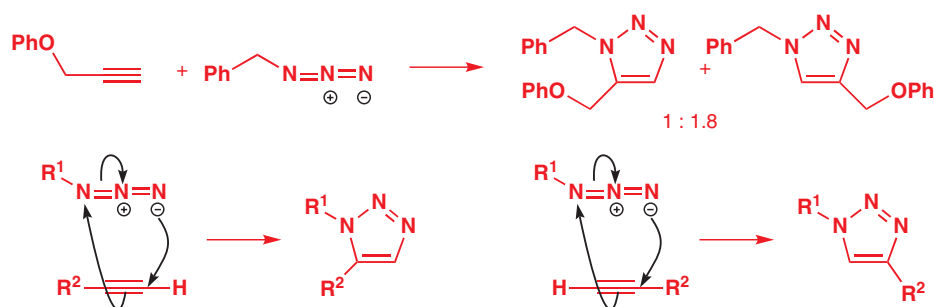
➡ The synthesis of indoles with this substitution pattern by the Fischer indole synthesis is described on p. 777.

The synthesis of the anti-inflammatory drug broperamole illustrates modification of a tetrazole using its anion. The tetrazole is again constructed from the nitrile—it's an aromatic nitrile with an electron-withdrawing substituent so this will be a good reaction.

Conjugate addition to acrylic acid (Chapter 22) occurs to give the other tautomer to the one we have drawn. The anion intermediate is, of course, delocalized and can react at any of the nitrogen atoms. Amide formation completes the synthesis of broperamole.

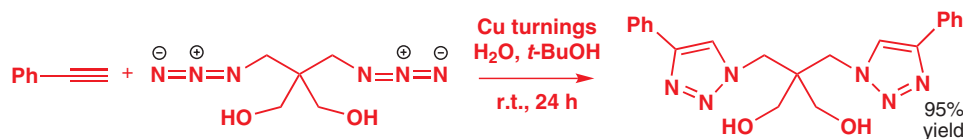


One of the best reactions of all is the related cycloaddition of a substituted azide to an alkyne. Just mixing together and heating an azide and an alkyne will give a triazole, but often as a mixture of two regioisomers.



 Interactive mechanism for triazole formation

However, a simple addition to the reaction mixture improves the situation hugely: a catalytic amount of Cu(I), often made *in situ* by adding CuSO₄ and a mild reducing agent, makes the reaction much faster and gives the 1,4-disubstituted triazole selectively. The work of Sharpless has turned this reaction into not only a very powerful way of making triazoles, but also a very simple way of linking two otherwise relatively unreactive molecules together—the reaction even works in water.

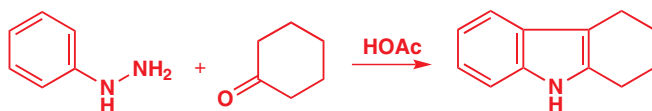


■ 1,2,4-Triazoles are usually made from the reaction of the unsubstituted 1,2,4-triazole anion with electrophiles, as described in Chapter 29.

■ It may look as though the more nucleophilic end of the azide has attacked the wrong end of the alkyne but you will see in Chapter 34 that (1) it is very difficult to predict which is the more nucleophilic end of a 1,3-dipole and (2) it may be either HOMO (dipole) and LUMO (alkyne), or LUMO (dipole) and HOMO (alkyne) that dominate the reaction.

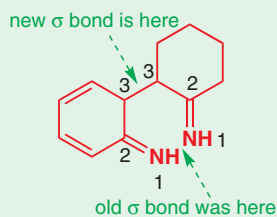
The Fischer indole synthesis

You are about to see one of the great inventions of organic chemistry. It is a remarkable reaction, amazing in its mechanism, and it was discovered in 1883 by one of the greatest organic chemists of all, Emil Fischer. Fischer had earlier discovered phenylhydrazine (PhNHNH₂) and, in its simplest form, the Fischer indole synthesis occurs when phenylhydrazine is heated in acidic solution with an aldehyde or ketone.



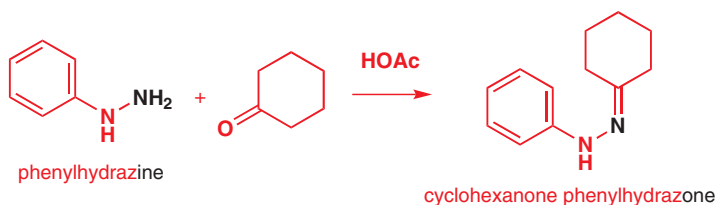
Emil Fischer (1852–1919), discovered phenylhydrazine as a PhD student in 1875, succeeded Hofmann at Berlin in 1900 where he built the then largest chemical institute in the world, and was awarded the Nobel prize in 1902. As well as his work on indoles, he laid the foundations of carbohydrate chemistry by completing the structure and synthesis of the main sugars. If only he hadn't also invented Fischer projections.

■ This step is a [3,3]-sigmatropic rearrangement, as you will discover in Chapter 35: the new single bond (C–C) bears a 3,3 relationship to the old single bond (N–N).

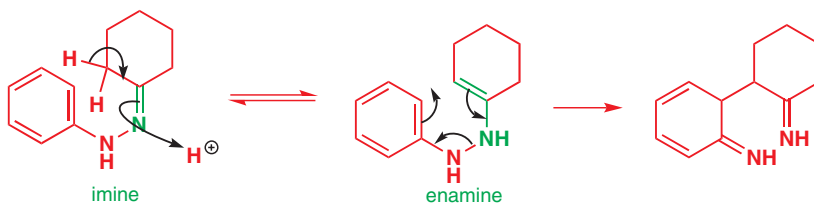


Interactive mechanism for Fischer indole synthesis

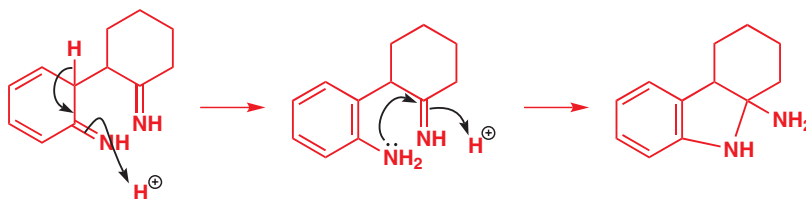
The first step in the mechanism is formation of the phenylhydrazone (the imine) of the ketone. This can be isolated as a stable compound (Chapter 11).



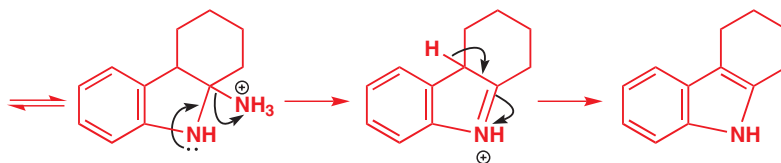
The hydrazone then needs to tautomerize to the enamine, and now comes the key step in the reaction. The enamine can rearrange with formation of a strong C–C bond and cleavage of the weak N–N single bond by moving electrons round a six-membered ring.



Next, re-aromatization of the benzene ring (by proton transfer from carbon to nitrogen) creates an aromatic amine that immediately attacks the other imine. This gives an aminal, the nitrogen equivalent of an acetal.

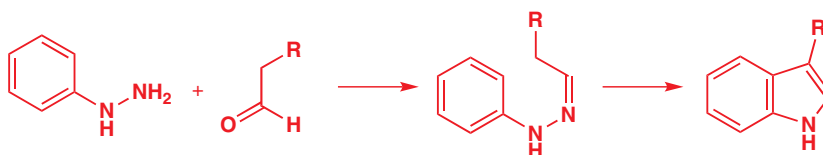


Finally, acid-catalysed decomposition of the aminal in acetal fashion with expulsion of ammonia allows the loss of a proton and the formation of the aromatic indole.

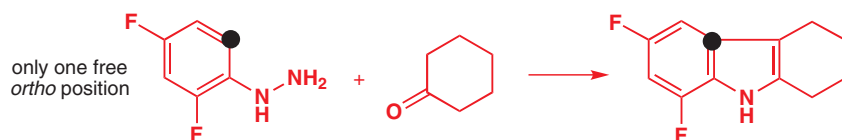


This is admittedly a complicated mechanism but if you remember the central step—the rearrangement of the enamine—the rest should fall into place. The key point is that the C–C bond is established at the expense of a weak N–N bond. Naturally, Fischer had no idea of any of the steps in the mechanism. He was sharp enough to see that something remarkable had happened and skilful enough to find out what it was.

The Fischer method is the main way of making indoles, but it is not suitable for them all. We need now to consider its applicability to various substitution patterns. If the carbonyl compound can enolize on one side only, as is the case with an aldehyde, then the obvious product is formed.

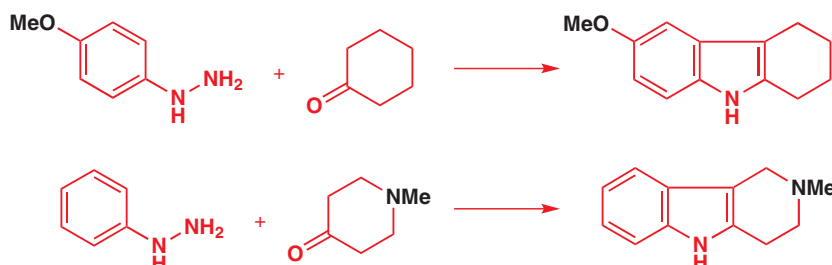


If the benzene ring has only one *ortho* position, then again cyclization must occur to that position. Other substituents on the ring are irrelevant.

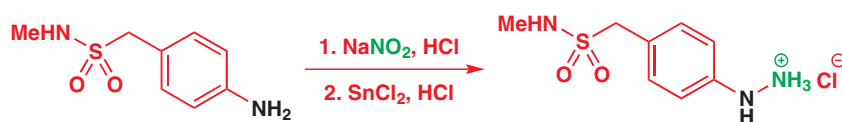
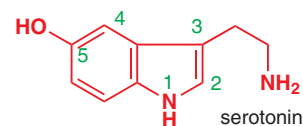


■ At this point we shall stop drawing the intermediate phenylhydrazone.

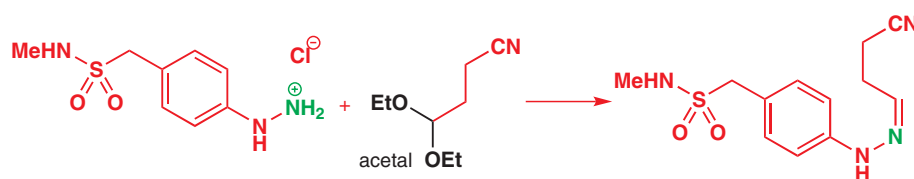
Another way to secure a single indole as product from the Fischer indole synthesis is to make sure the reagents are symmetrical. These two examples should make plain the types of indole available from symmetrical starting materials.



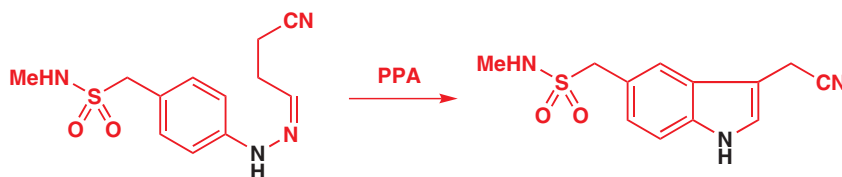
The substitution pattern of the first example is particularly important as the neurotransmitter serotonin is an indole with a hydroxyl group in the 5-position, and many important drugs follow that pattern. Sumatriptan (marketed as Imigran, the migraine treatment) is an analogue of serotonin, whose synthesis starts with the formation of a diazonium salt (Chapter 22) from the aniline shown below. Nitrosation gives the diazonium salt, and reduction with SnCl_2 and HCl returns the salt of the phenylhydrazine.



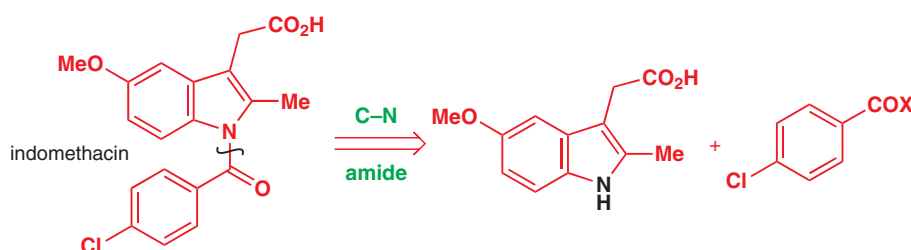
The required aldehyde (3-cyanopropanal) is added as an acetal to prevent self-condensation. The acidic conditions release the aldehyde, which forms the phenylhydrazone, ready for the next step.



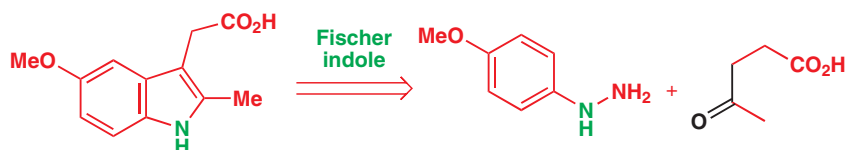
The Fischer indole synthesis itself is catalysed in this case by polyphosphoric acid (PPA), a sticky gum based on phosphoric acid (H_3PO_4) but dehydrated so that it contains some oligomers. It is often used as a catalyst in organic reactions and residues are easily removed in water.



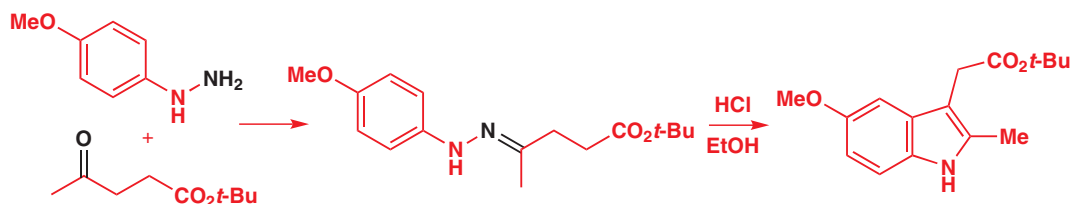
All that remains is to introduce the dimethylamino group. The nitrile is reduced by hydrogenation and the two methyl groups added by reductive amination with formaldehyde. The reducing agent is formic acid, and the reaction works by sequential formation



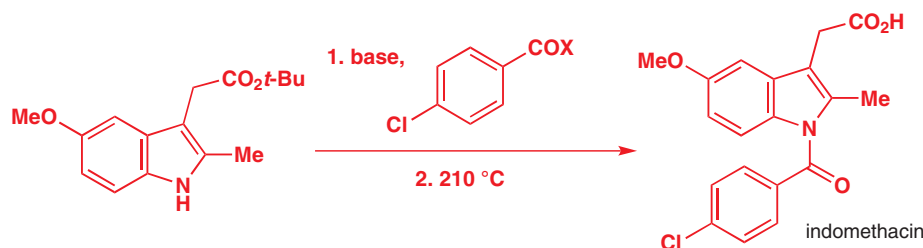
The benzene ring portion is symmetrical and is ideal for the Fischer synthesis but the right-hand half must come from an unsymmetrical open-chain keto-acid. Is it possible to control such a synthesis?



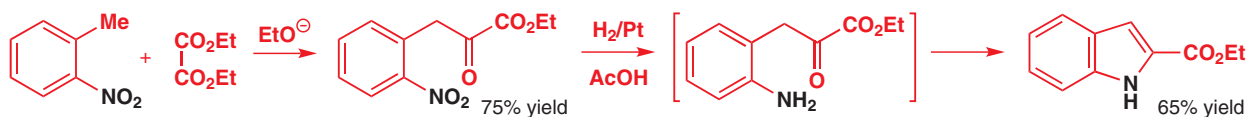
The Fischer indole is acid-catalysed so we must ask: on what side of the ketone is enolization (and therefore enamine formation) expected in acid solution? The answer (see Chapter 20) is away from the methyl group and into the alkyl chain. This is what we want and the reaction does indeed go this way. In fact, the *tert*-butyl ester is used instead of the free acid.



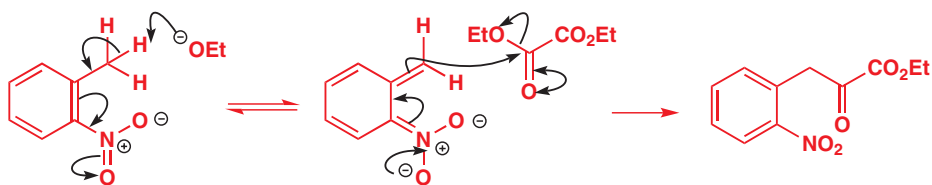
Acylation at the indole nitrogen atom is achieved with acid chloride in base and removal of the *t*-butyl ester gives free indomethacin.



There are many other indole syntheses but we will give a brief mention to only one other, which allows the synthesis of indoles with a different substitution pattern in the benzene ring. If you like names, you may call it the Reissert synthesis, and this is the basic reaction.

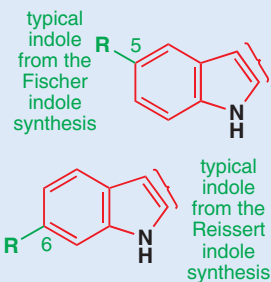


Ethoxide is a strong enough base to remove a proton from the methyl group, delocalizing the negative charge into the nitro group. The anion then attacks the reactive diester (diethyl oxalate) and is acylated by it.

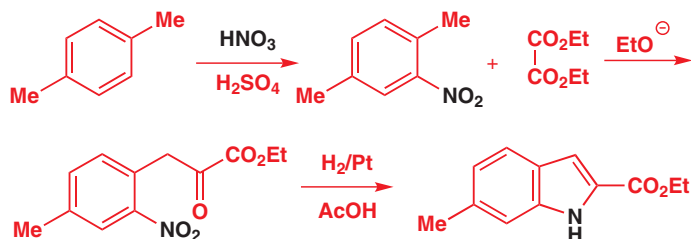


Fischer vs Reissert

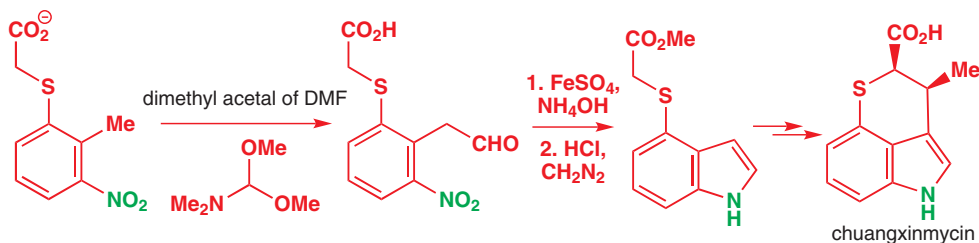
We can contrast the types of indole made by the Fischer and Reissert syntheses by the different ideal positions for substituents. These are, of course, not the only possible substitution patterns.



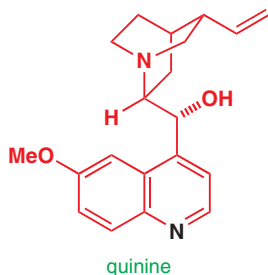
The rest of the synthesis is more straightforward: the nitro group can be reduced to an amine, which immediately forms an enamine by intramolecular attack on the more reactive carbonyl group (the ketone) to give the aromatic indole. Since the nitro compound is made by nitration of a benzene ring, the preferred symmetry is very different from that needed for the Fischer synthesis. Nitration of *para*-xylene (1,4-dimethylbenzene) is a good example.



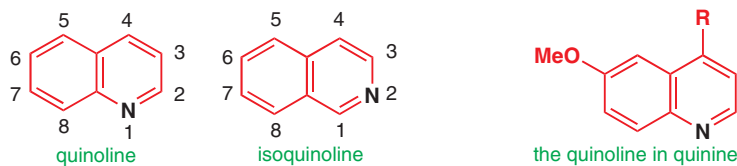
The ester products we have been using so far can be hydrolysed and decarboxylated by the mechanism described in the last chapter if a free indole is required. In any case, it is not necessary to use diethyl oxalate as the electrophilic carbonyl compound. The synthesis below, using the acetal of DMF as the electrophile, forms part of the synthesis of the strange antibiotic chuangxinmycin.



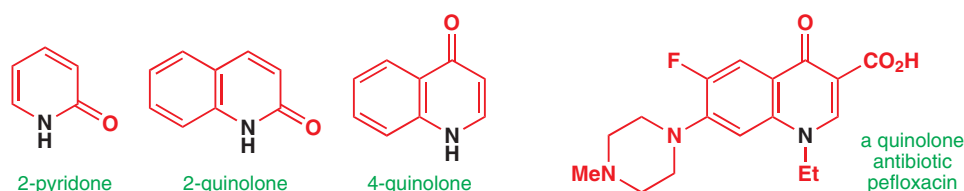
Quinolines and isoquinolines



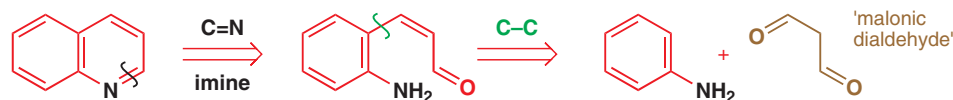
Quinoline forms part of the structure of quinine, the malaria remedy found in cinchona bark and known since the time of the Incas. The quinoline in quinine has a 6-MeO substituent and a side chain attached to C4. In discussing the synthesis of quinolines, we will be particularly interested in this pattern. This is because the search for anti-malarial compounds continues and other quinolines with similar structures are among the available anti-malarial drugs.



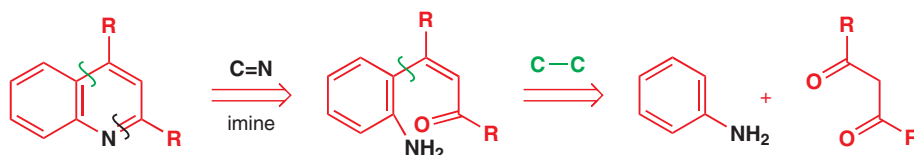
We shall also be very interested in quinolones, analogous to pyridones, with carbonyl groups at positions 2 and 4, as these are useful antibiotics. A simple example is pefloxacin, which has typical 6-F and 7-piperazine substituents.



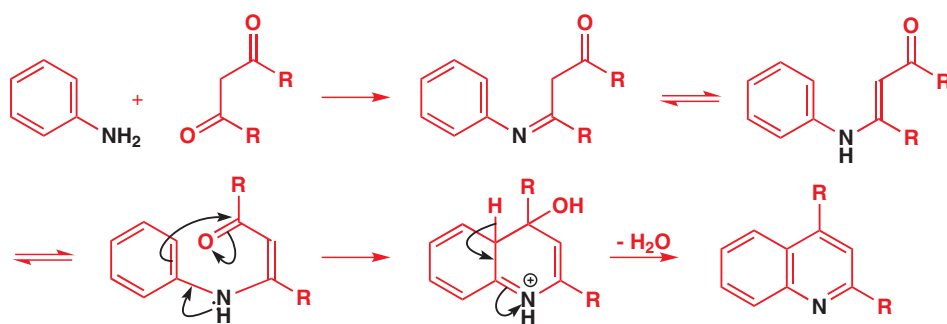
When we consider the synthesis of a quinoline, the obvious disconnections are, first, the C–N bond in the pyridine ring and, then, the C–C bond that joins the side chain to the benzene ring. We will need a three-carbon (C_3) synthon, electrophilic at both ends, which will yield two double bonds after incorporation. The obvious choice is a 1,3-dicarbonyl compound.



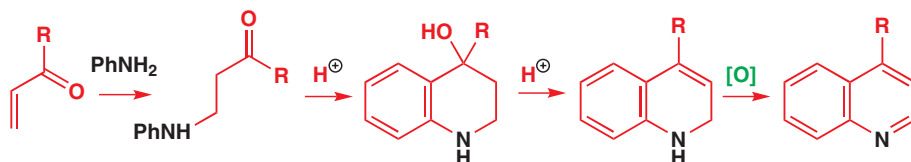
The choice of an aromatic amine is a good one as the NH_2 group reacts well with carbonyl compounds and it activates the *ortho* position to electrophilic attack. However, the dialdehyde is malonic dialdehyde, a compound that does not exist, so some alternative must be found. If the quinoline is substituted in the 2- and 4-positions this approach looks better.



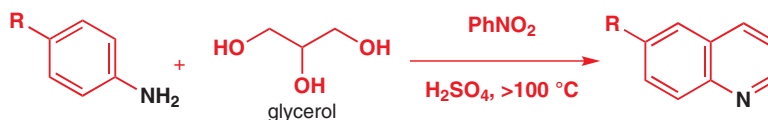
The initially formed imine will tautomerize to a conjugated enamine and cyclization now occurs by electrophilic aromatic substitution. The enamine will normally prefer to adopt the first configuration shown in which cyclization is not possible, and (perhaps for this reason or perhaps because it is difficult to predict which quinoline will be formed from an unsymmetrical 1,3-dicarbonyl compound) this has not proved a very important quinoline synthesis. However, the synthetic plan is sound, and we shall describe two important variants on this theme, one for quinolines and one for quinolones.



In the synthesis of pyridines it proved advantageous to make a dihydropyridine and oxidize it to a pyridine afterwards. The same idea works well in probably the most famous quinoline synthesis, the Skraup reaction. The diketone is replaced by an unsaturated carbonyl compound so that the quinoline is formed regioselectively. The first step is conjugate addition of the amine. Under acid catalysis the ketone now cyclizes in the way we have just described to give a dihydroquinoline after dehydration. Oxidation to the aromatic quinoline is an easy step accomplished by many possible oxidants.

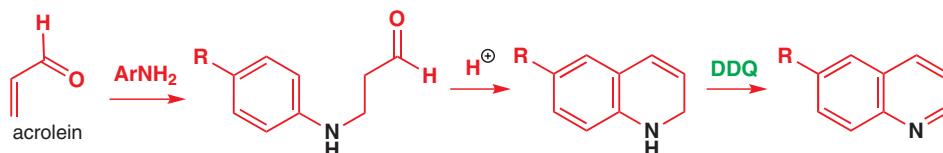


Traditionally, the Skraup reaction was carried out by mixing everything together and letting it rip. A typical mixture to make a quinoline without substituents on the pyridine ring would be the aromatic amine, concentrated sulfuric acid, glycerol, and nitrobenzene all heated up in a large flask at over 100 °C with a wide condenser.

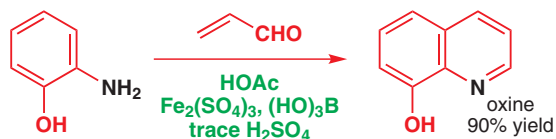


The ugly name of the Skraup reaction appropriately applies to the worst 'witch's brew' of all the heterocyclic syntheses. Some workers have added strange oxidizing agents such as arsenic acid, iron (III) salts, tin (IV) salts, nitrobenzenes of various substitution patterns, or iodine to make it 'go better'.

The glycerol was to provide acrolein ($\text{CH}_2=\text{CH}\cdot\text{CHO}$) by dehydration, the nitrobenzene was to act as oxidant, and the wide condenser...? All too often Skraup reactions did let rip—with destructive results. A safer approach is to prepare the conjugate adduct first, cyclize it in acid solution, and then oxidize it with one of the reagents we described for pyridine synthesis, particularly quinones such as DDQ.

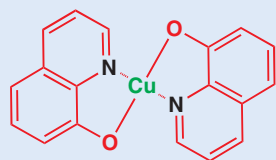


The more modern style of Skraup synthesis is used to make 8-quinolinol or 'oxine'. *ortho*-Aminophenol has only one free position *ortho* to the amino group and is very nucleophilic, so acrolein can be used in weak acid with only a trace of strong acid. Iron(III) is the oxidant, with a bit of boric acid for luck, and the yield is excellent.



Oxine

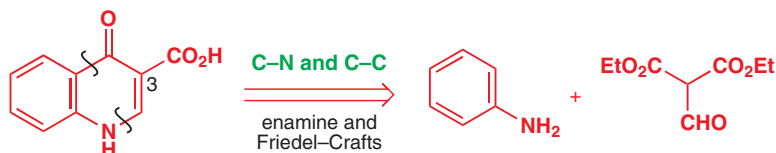
This compound is important because it forms unusually stable metal complexes with metal ions such as Mg(II) or Al(III). It is also used as a corrosion inhibitor on copper because it forms a stable layer of the Cu(II) complex that prevents oxidation of the interior.



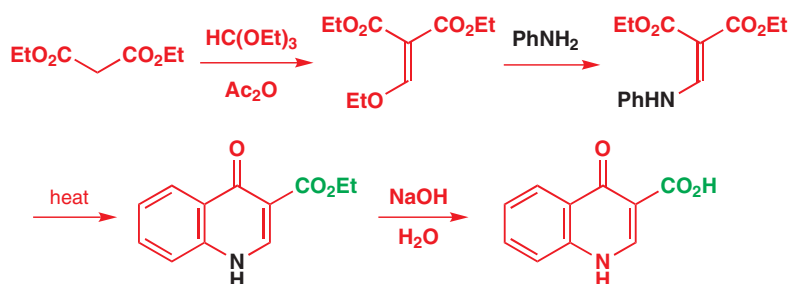
oxine complex of copper

Quinolones also come from anilines by cyclization to an *ortho* position

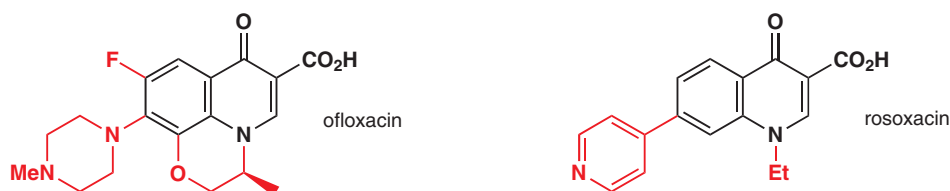
The usual method for making quinolone antibiotics is possible because they all have a carboxylic acid in the 3-position. The disconnection we used for quinoline suggests a rather unstable malonic ester derivative as starting material.



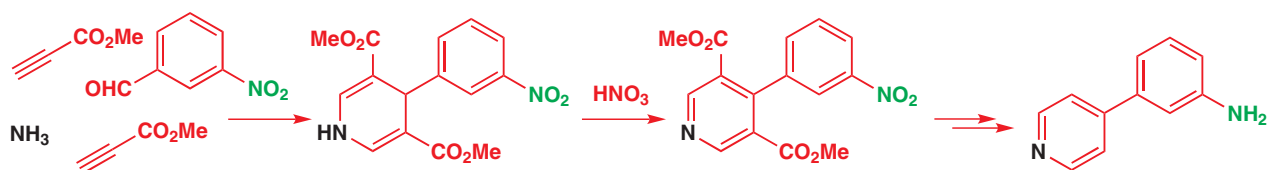
In fact, the enol ether of this compound is easily made from diethyl malonate and ethyl orthoformate [$\text{HC}(\text{OEt})_3$]. The aromatic amine reacts with this compound by an addition-elimination sequence, giving an enamine that cyclizes on heating. This time the geometry of the enamine is not a concern.



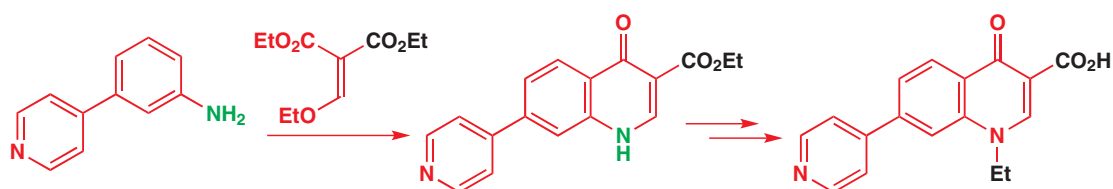
For examples of quinolone antibiotics we can choose ofloxacin, whose synthesis was discussed in detail in Chapter 22, and rosoxacin, whose synthesis is discussed below. Both molecules contain the same quinolone carboxylic acid framework, outlined in black, with another heterocyclic system at position 7 and various other substituents here and there.



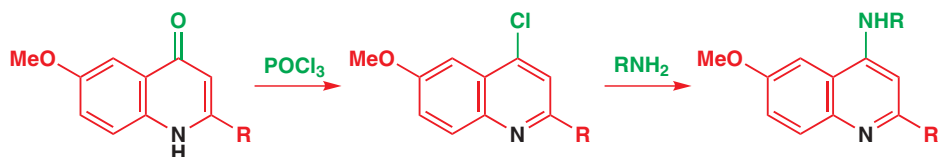
To make rosoxacin two heterocyclic systems must be constructed. Workers at the pharmaceutical company Sterling decided to build the pyridine in an ingenious version of the Hantzsch synthesis using acetylenic esters on 3-nitrobenzaldehyde. The ammonia was added as ammonium acetate. Oxidation with nitric acid made the pyridine, hydrolysis of the esters and decarboxylation removed the acid groups, and reduction with Fe(II) and HCl converted the nitro group into the amino group required for the quinolone synthesis.



Now the quinolone synthesis can be executed with the same reagents we used before and all that remains is ester hydrolysis and alkylation at nitrogen. Notice that the quinolone cyclization could in theory have occurred in two ways as the two positions *ortho* to the amino group are different. In practice cyclization occurs away from the pyridine ring as the alternative quinolone would be impossibly crowded.



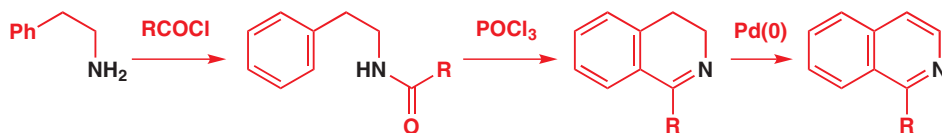
Since quinolones, like pyridones, can be converted into chloro-compounds with POCl_3 , they can be used in nucleophilic substitution reactions to build up more complex quinolines.



■ The Vilsmeier reaction with DMF is on p. 734.

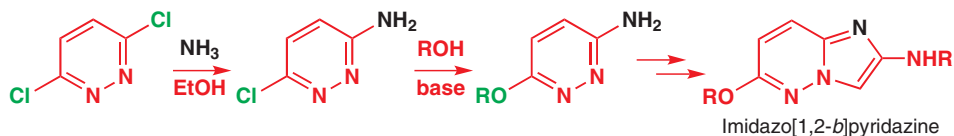
■ The reaction with Pd is simply the reverse of a Pd-catalysed hydrogenation.

We will give just one important synthesis of isoquinolines here. It is a synthesis of a dihydroisoquinoline by what amounts to an intramolecular Vilsmeier reaction in which the electrophile is made from an amide and POCl_3 . Oxidation (in this case a dehydrogenation with $\text{Pd}(0)$) gives the quinoline.

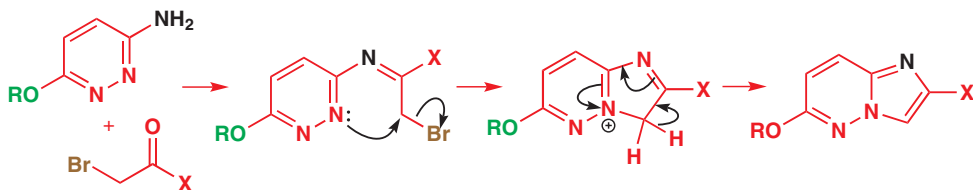


More heteroatoms in fused rings mean more choice in synthesis

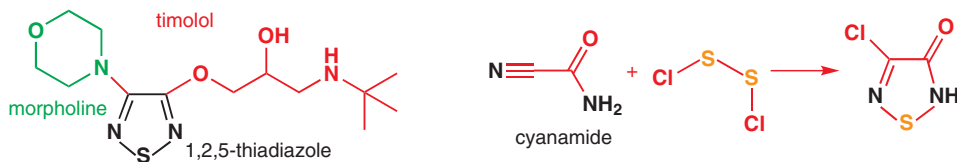
The imidazo-pyridazine ring system forms the basis for a number of drugs in human and animal medicine. The synthesis of this system uses the chemistry discussed in Chapter 29 to build the pyridazine ring. There we established that it was easy to make dichloropyridazines and to displace the chlorine atoms one by one with different nucleophiles. Now we will move on from these intermediates to the bicyclic system.



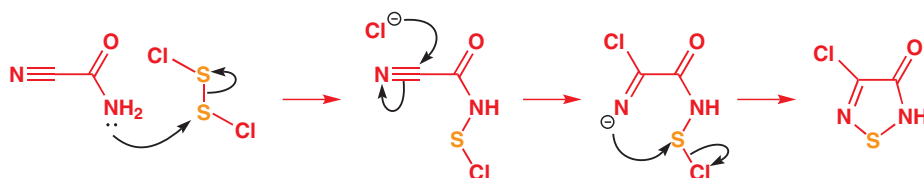
A 2-bromo-acid derivative is the vital reagent. It reacts at the amino nitrogen atom with the carbonyl group and at the pyridazine ring nitrogen atom with the alkyl halide. This is the only way the molecule can organize itself into a ten-electron aromatic system.



In Chapter 29 we also gave the structure of timolol, a thiadiazole-based β -blocker drug for reduction of high blood pressure. This compound has an aromatic 1,2,5-thiadiazole ring system and a saturated morpholine as well as an aliphatic side chain. Its synthesis relies on ring formation by rather a curious method followed by selective nucleophilic substitution, rather in the style of the last synthesis. The aromatic ring is made by the action of S_2Cl_2 on 'cyanamide'.

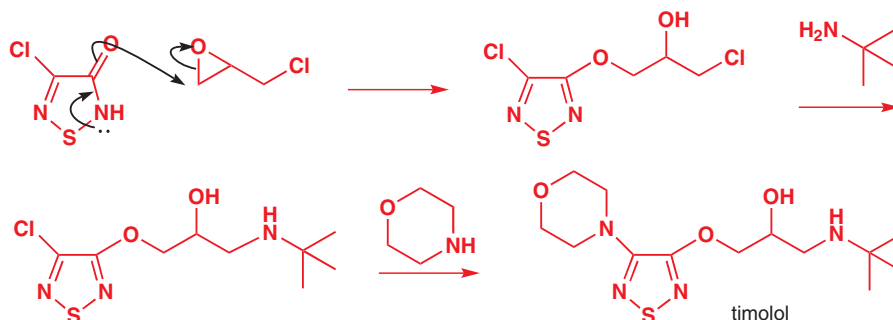


This reaction must start by attack of the amide nitrogen on the electrophilic sulfur atom. Cyclization cannot occur while the linear nitrile is in place so chloride ion (from disproportionation of $\text{ClS}^\cdot$) must first attack CN. Thereafter cyclization is easy.



Reaction with epichlorohydrin followed by amine displacement puts in one of the side chains and nucleophilic substitution with morpholine on the ring completes the synthesis.

➡ We used epichlorohydrin a lot in Chapter 28.



Summary: the three major approaches to the synthesis of aromatic heterocycles

We end this chapter with summaries of the three major strategies in the synthesis of heterocycles:

- ring construction by ionic reactions
- ring construction by cycloadditions
- modification of existing rings by electrophilic or nucleophilic aromatic substitution or by lithiation and reaction with electrophiles.

We will summarize the different applications of these strategies, and also suggest cases for which each strategy is not suitable. This section revises material from Chapter 29 as well since most of the ring modifications appear there.

■ This is only a summary. There are more details in the relevant sections of Chapters 29 and 30. There are also many, many more ways of making all these heterocycles. These methods are just where we suggest you *start*.

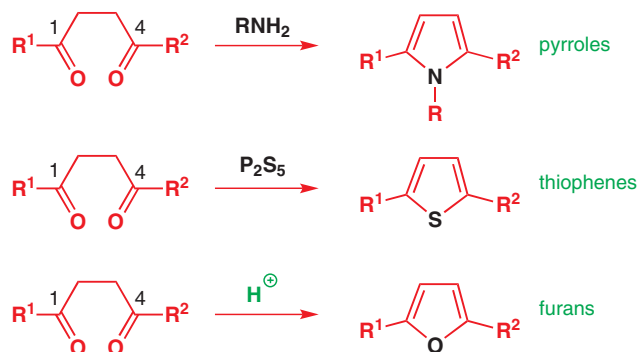
Ring construction by ionic cyclization

The first strategy you should try out when faced with the synthesis of an aromatic heterocyclic ring is the disconnection of bonds between the heteroatom or atoms and carbon, with the idea of using the heteroatoms as nucleophiles and the carbon fragment as a double electrophile.

Heterocycles with one heteroatom

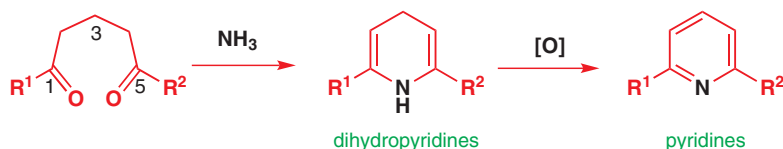
five-membered rings

- pyrroles, thiophenes, and furans ideally made by this strategy from 1,4-dicarbonyl compounds

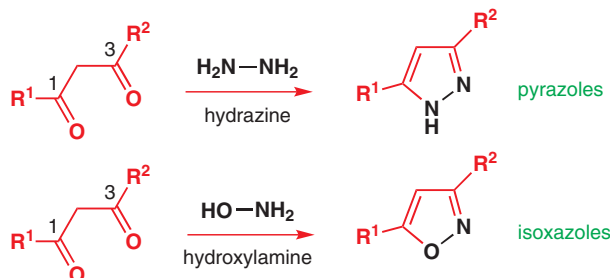


six-membered rings

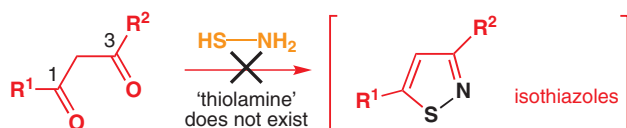
- pyridines made by this strategy from 1,5-dicarbonyl compounds with oxidation

**Heterocycles with two adjacent heteroatoms***five-membered rings*

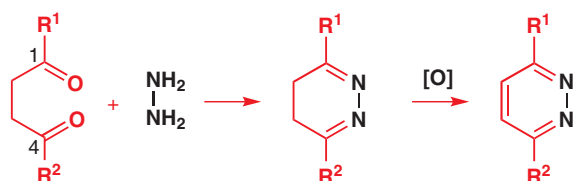
- pyrazoles and isoxazoles ideally made by this strategy from 1,3-dicarbonyl compounds



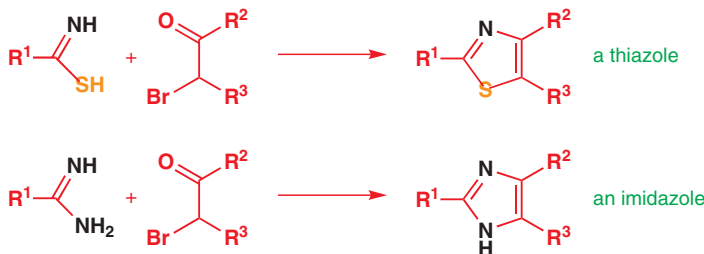
Note. This strategy is *not* suitable for isothiazoles as 'thiolamine' does not exist

*six-membered rings*

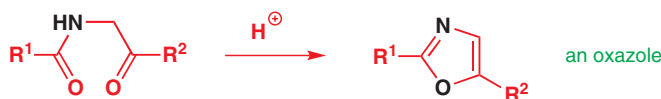
- pyridazines ideally made by this strategy from 1,4-dicarbonyl compounds with oxidation

**Heterocycles with two non-adjacent heteroatoms***five-membered rings*

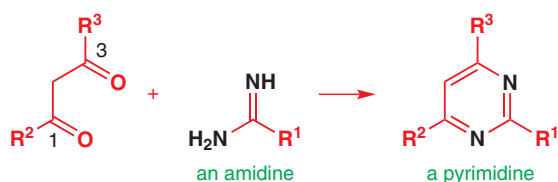
- imidazoles and thiazoles ideally made by this strategy from α -halocarbonyl compound



Note. This strategy is *not* suitable for oxazoles as amides are not usually reactive enough: cyclization of acylated carbonyl compounds is usually preferred

*six-membered rings*

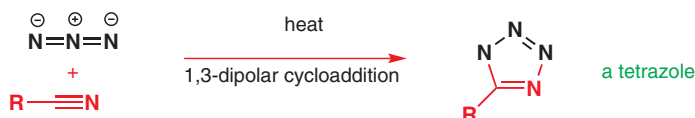
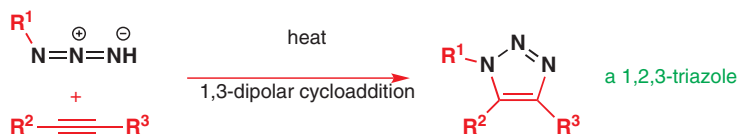
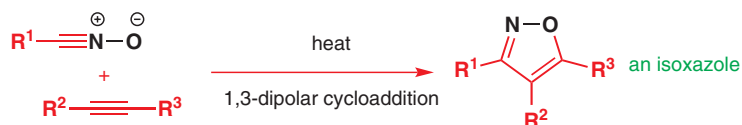
- pyrimidines ideally made by this strategy from 1,3-dicarbonyl compounds



Ring construction by cycloadditions

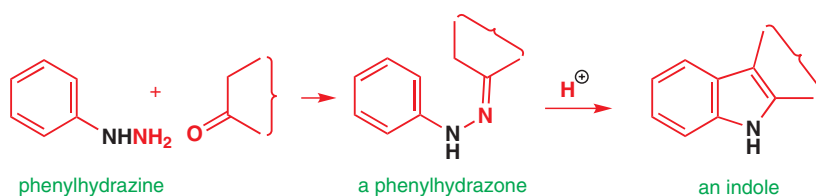
1,3-dipolar cycloaddition reactions

- ideal for the construction of isoxazoles, 1,2,3-triazoles, and tetrazoles



...or sigmatropic rearrangements

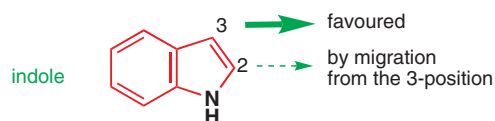
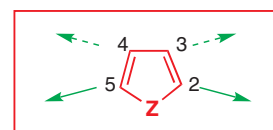
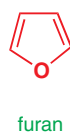
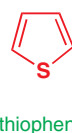
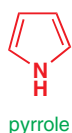
- a special reaction that is the vital step of the Fischer indole synthesis



Ring modification

Electrophilic aromatic substitution

- works very well on pyrroles, thiophenes, and furans, where it occurs best in the 2- and 5-positions and nearly as well in the 3- and 4-positions
- often best to block positions where substitution not wanted
- works well for indole—occurs only in the 3-position but the electrophile may migrate to the 2-position



- works well for five-membered rings with a sulfur, oxygen, or pyrrole-like nitrogen atom and occurs anywhere that is not blocked (see earlier sections)

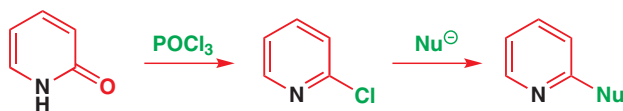
Note. Not recommended for pyridine, quinoline, or isoquinoline

Nucleophilic aromatic substitution

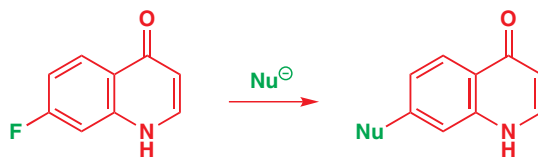
- works particularly well for pyridine and quinoline where the charge in the intermediate can rest on nitrogen



- especially important for pyridones and quinolones with conversion to the chloro-compound and displacement of chlorine by nucleophiles and, for quinolines, displacement of fluorine atoms on the benzene ring

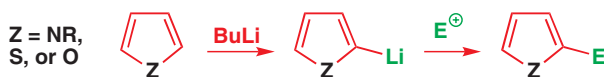


- works well for the six-membered rings with two nitrogens (pyridazines, pyrimidines, and piperazines) in all positions



Lithiation and reaction with electrophiles

- works well for pyrrole (if NH blocked), thiophene, or furan next to the heteroatom. Exchange of Br or I for Li works well for most electrophiles providing any acidic hydrogens (including the NH in the ring) are blocked



Further reading

The best general text on heterocycles is J. A. Joule and K. Mills, *Heterocyclic Chemistry* 4th edn, Chapman and Hall, London, 2010.

S. Warren and P. Wyatt, *Workbook for Organic Synthesis: the Disconnection Approach*, Wiley, Chichester, 2009, chapters 34–35.

Check your understanding



To check that you have mastered the concepts presented in this chapter, attempt the problems that are available in the book's Online Resource Centre at <http://www.oxfordtextbooks.co.uk/orc/clayden2e/>